AGILE

PROJECT MGMT

TRAINER SLIDES · WSQ

Agile Project Management for Business

WSQ Course Code: TGS-2023018967

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

2 days · 16 training hours · TSC: Business Agility (ICT-BIN-4038-1.1)

Version v9.0 · 17 August 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

WSQ Adult Educator · Agile Practitioner

NAME

CERTIFICATIONS

INDUSTRY EXPERIENCE

AGILE EXPERIENCE

CONTACT

About the Trainer

AA

Dr Alfred Ang

PhD (NUS) · MEng (NTU) · MBA · ACTA/DACE certified adult educator

ROLE
Course developer and principal trainer, Tertiary Infotech Academy

CERTIFICATIONS
Enterprise Design Thinking Practitioner · Agile & Scrum certified · ACTA/DACE

EXPERIENCE
20+ years across engineering, data analytics, AI and digital transformation

TEACHES
Agile project management, design thinking, problem solving and analytics

CONTACT
enquiry@tertiaryinfotech.com · +65 6100 0613

Ground Rules

1. **Phones on silent**
   Set your mobile phone to silent mode for the duration of the session.

2. **Participate actively**
   No question is a stupid question. This course runs on discussion.

3. **Mutual respect**
   Agree to disagree. One conversation at a time.

4. **Be punctual**
   Return from breaks on time so we finish on schedule.

5. **Step out quietly**
   Exit silently for a phone call or a toilet break.

6. **75% attendance**
   Required to be eligible for assessment and funding.

Download Your Course Material

https://lms-tms.tertiaryinfotech.com

Log in with the email you registered with

My Courses select Agile Project Management for Business

Courseware Learner Slides · Learner Guide · Activities

Download save the PDF to your laptop before we start

1. Open lms-tms.tertiaryinfotech.com in your browser
2. Log in with your registered email address
3. Open this course under 'My Courses'
4. Download the Learner Guide and slides
5. Keep them open — the assessment is open book

Everything you need for the activities and the open-book assessment is in the Learner Guide.

BY THE END OF THIS COURSE YOU WILL BE ABLE TO

Learning Outcomes

1. **LO1 — Adopt the Agile mindset**
   Adopt a new Agile mindset for project management, and know when Agile fits and when it does not.

2. **LO2 — Share and implement**
   Share and implement Agile practices within your teams — the Manifesto, Scrum, Lean and Kanban.

3. **LO3 — Build, execute and track**
   Build an Agile team to execute and track project performance with real metrics.

Skills Framework — TSC Abilities Assessed

Code	Ability	Where it is taught
A1	Share information actively within and across teams	Topic 2 · Activity 4
A2	Organise work in alignment with operational priorities	Topic 1–2 · Activities 2, 4
A3	Implement Agile or lean practices to reduce waste and defects	Topic 2–3 · Activities 3, 7
A4	Measure progress against targets on a regular basis	Topic 3 · Activities 5, 7, 8
A5	Experiment with new ideas, products or services	Topic 1 · Activity 1
A6	Assess work performance and quality for continuous improvement	Topic 3 · Activities 6, 7, 8
A7	Manage responsibilities and take ownership of outcomes	Topic 3 · Activities 5, 6, 8

Every ability is taught AND practised in a named activity — nothing is assessed that is not taught.

Skills Framework — TSC Knowledge Assessed

Code	Knowledge	Where it is taught
K1	Methods to analyse current and future business operating landscapes	Topic 1 · Activity 2
K2	Methods to analyse current and future customer needs and preferences	Topic 1 · Activity 1
K3	Organisational policies, processes and standards	Topic 1 · Activity 2
K4	Types of change management methodologies, tools and practices	Topic 2
K5	Types of team composition and formation models	Topic 3 · Activities 4, 5
K6	Values and principles of Agile methodologies	Topic 2 · Activity 3
K7	Types of Agile methodologies and practices	Topic 2 · Activity 3

Every knowledge code is taught in a topic AND tested by one question in the Written Assessment.

Lesson Plan — Day 1

1. **09:30 Welcome & admin**
   Digital attendance (AM) · introductions · ground rules · outcomes

2. **11:00 Topic 1**
   Introduction to Agile Project Management (Activities 1–2)

3. **13:00 Lunch break**
   1 hour · digital attendance (PM) on return

4. **16:45 Topic 2**
   Agile Essentials begins (Activities 3–4 on Day 2)

5. **18:30 End of Day 1**
   Recap and Q&A

Lesson Plan — Day 2

1. **09:30 Digital attendance (AM)**
   Recap of Day 1
2. **09:45 Topic 2 continued**
   Lean · Kanban · XP · change management (Activities 3–4)
3. **13:00 Lunch break**
   1 hour · digital attendance (PM) on return
4. **14:45 Topic 3**
   Agile Project Execution and Tracking (Activities 5–8)
5. **18:30 Feedback & TRAQOM**
   Course feedback and the mandatory TRAQOM survey
6. **19:00 Final Assessment**
   Digital attendance (Assessment) · WA (1 h) + Case Study (1 h)

WHAT WE COVER OVER 2 DAYS · 16 TRAINING HOURS

Course Outline

1

Topic 1 — Introduction to Agile Project Management

Business landscape · Waterfall · Agile overview · The paradigm shift · K1, K2, K3, A2, A5

2

Topic 2 — Agile Essentials

Manifesto values & principles · Scrum · Lean · Kanban · XP · Overcoming resistance · K4, K6, K7, A1, A3

3

Topic 3 — Agile Project Execution and Tracking

Team building & vision · User stories & estimation · Execution · Metrics · Continuous improvement · K5, A4, A6, A7

HANDS-ON TOOLING · BROWSER-BASED, NOTHING TO INSTALL

The Six Tools You Will Use

1 5 Whys

Root-cause analysis in the sprint retrospective (Activity 6)
https://alfredang.github.io/5whys/

2 Fishbone

Cause-and-effect analysis of delivery problems (Activity 2)
https://alfredang.github.io/fishbone/

3 Pareto Chart

Prioritising the 20% of causes driving 80% of defects (Activity 7)
https://alfredang.github.io/paretochart/

4 RACI Matrix

Clarifying Agile role accountability (Activity 4)
https://alfredang.github.io/raci/

5 Scrum Board

Running the sprint board, backlog and burndown (Activities 3, 5, 8)
https://alfredang.github.io/scrum/

6 Design Thinking

Empathising with customers to shape the product vision (Activity 1)
https://alfredang.github.io/designthinking/

HOW WE WORK

Every activity uses a real browser-based tool on a single running case study — HarbourFront Logistics, a Singapore 3PL whose customer portal failed as a waterfall project.

THE RUNNING CASE

One case study runs through all 8 activities.

HarbourFront Logistics Pte Ltd — a 140-staff Singapore third-party logistics provider whose 'CustomerConnect' portal shipped 4 months late with 62% of its features unused. You will diagnose it, restart it with Agile, run its sprints, and forecast its release.

Briefing for Assessment

1 **Clear your table**
Place phones and other materials under the table or on the floor.

2 **No photos or recording**
Assessment scripts must not be photographed or recorded.

3 **No discussion**
Work individually once the assessment starts.

4 **Black or blue pen**
Use a black or blue pen for hard-copy scripts.

5 **No correction fluid**
Do not use liquid paper or correction tape.

6 **Scripts collected**
Scripts are collected when the time is up.

Final Assessment

1. **Written Assessment (WA)**
   Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.

2. **Case Study (CS)**
   Case Study (CS) — an applied business scenario with structured tasks, 1 hour, open book.

3. **Open book**
   Open book: slides, Learner Guide and approved course materials only.

4. **Competency**
   A minimum of 75% attendance is required to be eligible for assessment and funding. Learners must be assessed Competent in both instruments.

ASSESSMENT · END-TO-END FLOW

Assessment Flow

1. **BRIEFING** — Rules, timing, open-book scope
2. **WRITTEN (WA)** — Short-answer questions · 1 hour
3. **CASE STUDY (CS)** — Applied business scenario · 1 hour
4. **MARKING** — Assessor marks against the key
5. **RESULT & APPEAL** — Competent / NYC · feedback given

Both instruments must be assessed COMPETENT · Open book: slides, Learner Guide and approved materials only · 75% attendance required

Briefing → Written Assessment → Case Study → marking → result, with an appeal route at every stage.

Criteria for Funding

1. **75% attendance**
   Minimum attendance rate of 75%, based on the SSG digital attendance record.

2. **Assessed Competent**
   Complete both assessment instruments and be assessed as 'Competent'.

3. **TRAQOM survey**
   Complete the mandatory course feedback and TRAQOM survey.

01

TOPIC 01

Introduction to Agile Project Management

Business landscape · Waterfall · Agile overview · The paradigm shift · LO1 · K1, K2, K3, A2, A5

TOPIC 01 · THE CENTRAL QUESTION

Are we building the right thing?

Traditional project management asks whether we are building the thing right — on time, on budget, to specification. Agile asks the prior question first, and keeps asking it.

The Traditional Project Lifecycle

1	2	3	4	5
Initiate	**Plan**	**Execute**	**Monitor**	**Close**
charter, sponsor, business case	scope, WBS, schedule, budget	build to the approved plan	variance against the baseline	handover and lessons learned

Nothing here is wrong. It is simply built on an assumption: that the plan written at the start remains a good description of reality until the end.

The Challenges Every Project Manager Recognises

1. **Scope creep**
   Requirements change after sign-off, and every change costs a variation order.

2. **Resource constraints**
   Not enough people, or not enough of the right people at the right time.

3. **Unrealistic schedules**
   Dates set by commitment or hope rather than by capacity.

4. **Unproven technology**
   Platforms chosen early and validated far too late.

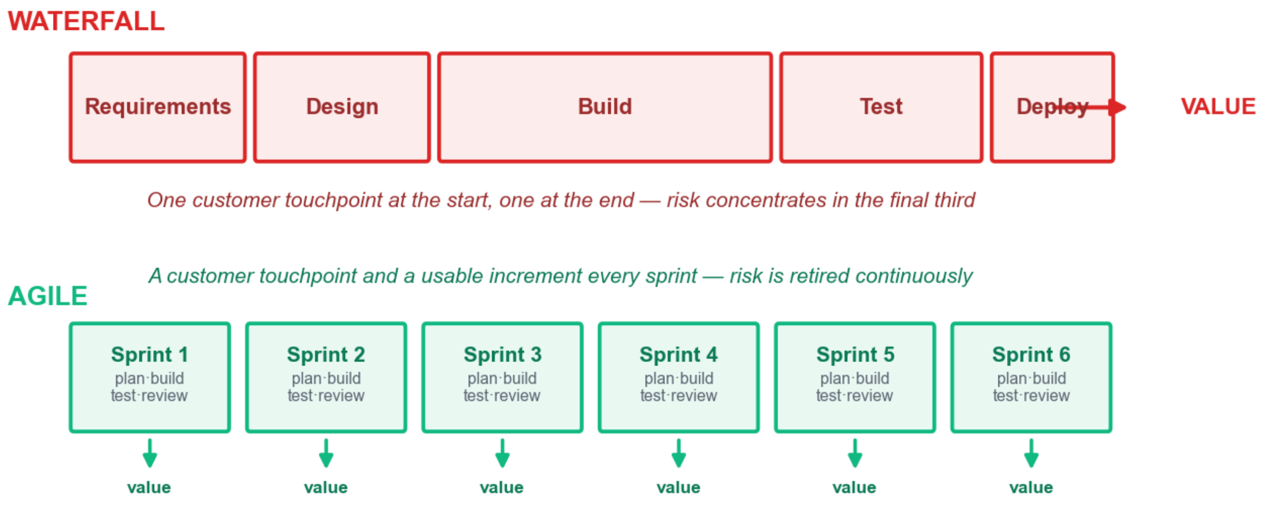
5. **Documentation burden**
   Reporting effort that grows faster than delivery effort.

6. **Too many dependencies**
   One late input stalls a whole chain of work.

7. **Stakeholder expectations**
   Different stakeholders holding different pictures of 'done'.

8. **Late discovery**
   The biggest problems surface in the final third, when change is most expensive.

Understanding the Waterfall Model



Scope frozen at the start

The end product is fixed in all respects when the project begins.

Detailed upfront estimation

Requires complete estimation and planning before any build.

Change is a cost event

Mid-course changes cause delays and formal variation orders.

One integration point

The parts meet late, so integration risk peaks at the worst moment.

Challenges with the Waterfall Model

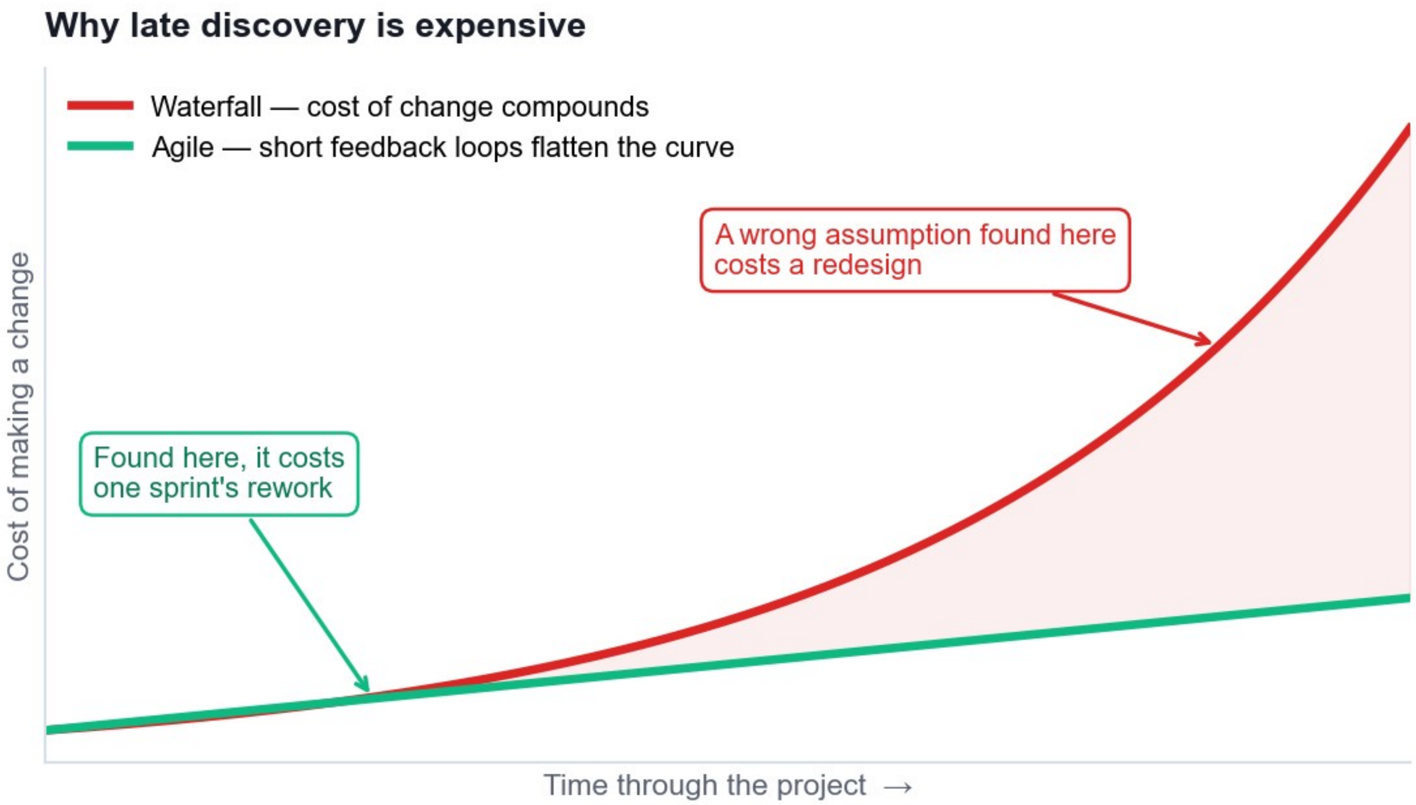
1 **Change is hostile**
Managing changing requirements mid-lifecycle fights the model rather than using it.

2 **Late visibility**
Nobody sees the end product until the end of the lifecycle.

3 **Risk loading**
Risk and uncertainty concentrate at the point where they are most expensive to fix.

4 **Complexity upfront**
The model asks you to fully understand a complex problem before you have built anything.

The Cost of Change



This single curve is the economic argument for short iterations — it is not about speed, it is about the price of being wrong.

What Agile Actually Is

1 An umbrella term

Agile covers many iterative, incremental methods — it is not one process.

2 Iterative delivery

APM: 'an iterative approach to delivering a project throughout its life cycle'.

3 Built for software, used everywhere

Now standard in marketing, finance, HR, construction and biotech.

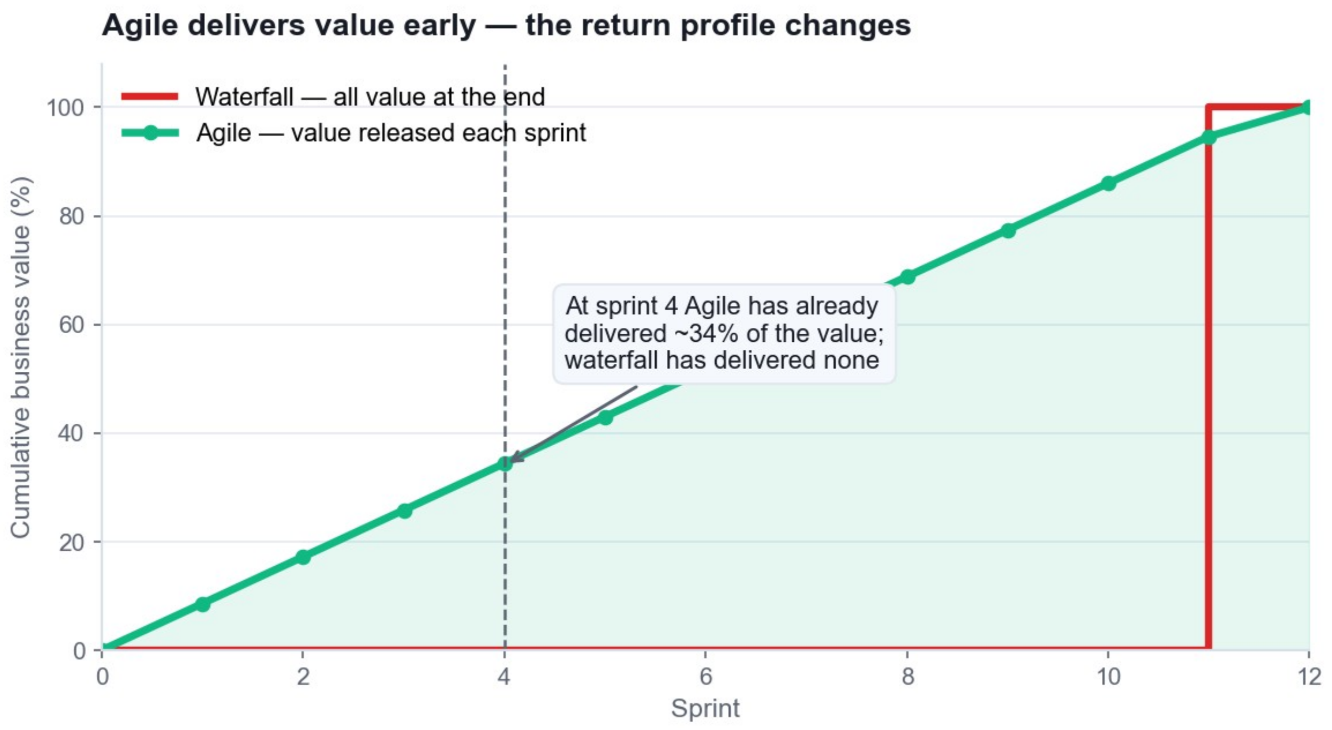
4 A mindset, not a ceremony

Running stand-ups without changing how decisions are made is not Agile.

THE CORE IDEA

Deliver in small increments, get feedback from real users, and let what you learn change what you build next. Everything else in this course is machinery for doing that reliably.

Agile vs Traditional Project Management



Builds in increments

Rather than delivering the whole product as one event.

Plans continuously

Planning happens throughout, not only at the start.

Customer sees value faster

Benefit is released during the project, not after it.

Change is welcomed

Change is expected and priced in, not resisted.

TOPIC 01 · SIDE BY SIDE

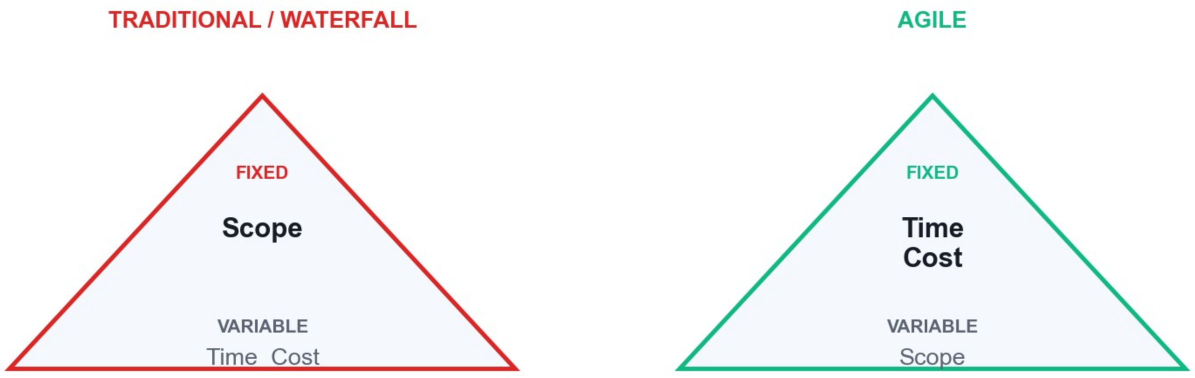
Comparison of Agile and Waterfall

Aspect	Waterfall / Traditional	Agile
Approach	Sequential phases, one pass	Iterative and incremental
Requirements	Fixed and signed off upfront	Emerge and are re-ordered continuously
Flexibility	Resists change after planning	Adapts throughout
Customer role	Consulted at start and at UAT	Continuous collaboration
Delivery	One release at the end	A usable increment every sprint
Risk	Addressed upfront, realised late	Retired continuously
Documentation	Comprehensive by mandate	Barely sufficient, just in time
Team structure	Hierarchical, specialised roles	Self-organising, cross-functional

Neither column is 'better'. The right question is which column matches the uncertainty of YOUR work.

Inverting the Triangle

Inverting the triangle: Agile fixes time and cost, and varies scope



Waterfall fixes scope

Time and cost flex to deliver the agreed feature list.

Agile fixes time and cost

You buy a stable team and a timebox; scope is the variable.

This is the real trade

You cannot fix all three and still respond to what you learn.

Implication for governance

Funding shifts from one big approval to incremental funding decisions.

Analysing the Business Operating Landscape (K1)

1

PESTLE

Political, economic, social, technological, legal and environmental forces reshaping the market.

2

SWOT

Internal strengths and weaknesses against external opportunities and threats.

3

Porter's Five Forces

Competitive rivalry, supplier and buyer power, substitutes and new entrants.

4

Scenario planning

Several plausible futures rather than one forecast, each with a response.

5

VUCA framing

Volatility, uncertainty, complexity and ambiguity — name which one you actually face.

6

Re-run the analysis

The Agile difference: quarterly, not once per project.

Analysing Customer Needs and Preferences (K2)

1. **Personas & empathy maps**
   What the customer says, thinks, does and feels — used in Activity 1.

2. **Journey maps**
   The end-to-end experience, exposing where the pain actually sits.

3. **Jobs to be done**
   The progress the customer is trying to make, not the feature they requested.

4. **Kano analysis**
   Sorting delighters, satisfiers, must-haves and indifferent features.

5. **A/B experiments**
   Testing a preference with real behaviour instead of asking an opinion.

6. **Continuous feedback**
   Sprint reviews turn customer contact into a rhythm, not an event.

Agile Beyond Software — Business Use Cases

1. **Adobe · marketing**
   Campaign teams run sprints and retrospectives to adapt messaging to live data.

2. **Toyota · manufacturing**
   The origin of Lean and Kanban — flow, WIP limits and waste elimination.

3. **Spotify · product delivery**
   Autonomous squads optimised for delivery speed and learning.

4. **PayPal · workforce alignment**
   Agile used to align large distributed teams on shared outcomes.

5. **Banking · compliance delivery**
   Hybrid Agile inside regulated stage-gate governance.

6. **Construction & biotech**
   Iterative design and staged validation where rework is costly.

The Agile Mindset

1. Welcoming change rather than resisting it
2. Working in small increments of real value
3. Using build-and-feedback loops to learn
4. Learning through discovery instead of assumption
5. Value-driven development, not activity-driven
6. Failing fast, and extracting the lesson
7. Continuous delivery of working outcomes
8. Continuous improvement of how the team works

TOPIC 01 · INTELLECTUAL HONESTY

When Agile Is NOT the Right Answer

1 Truly fixed scope

A regulatory submission with a legislated, unchangeable specification.

2 No customer access

If nobody can give feedback each sprint, the feedback loop is theatre.

3 Physical irreversibility

Where an increment cannot be built and changed cheaply.

4 No mandate to change governance

Agile teams inside stage-gate funding produce reports, not agility.

CHOOSE DELIBERATELY

Stable requirements and a documentation mandate favour waterfall; evolving requirements and high novelty favour Agile. Most real organisations run a considered hybrid.

WHAT YOU'LL DO

Hands-On Activities — Introduction to Agile Project Management

1 **Activity 1 · Empathise with the Customer to Reframe a Failing Project**

Tool: Design Thinking
45 minutes · LO1

2 **Activity 2 · Diagnose the Waterfall Failure with a Fishbone Analysis**

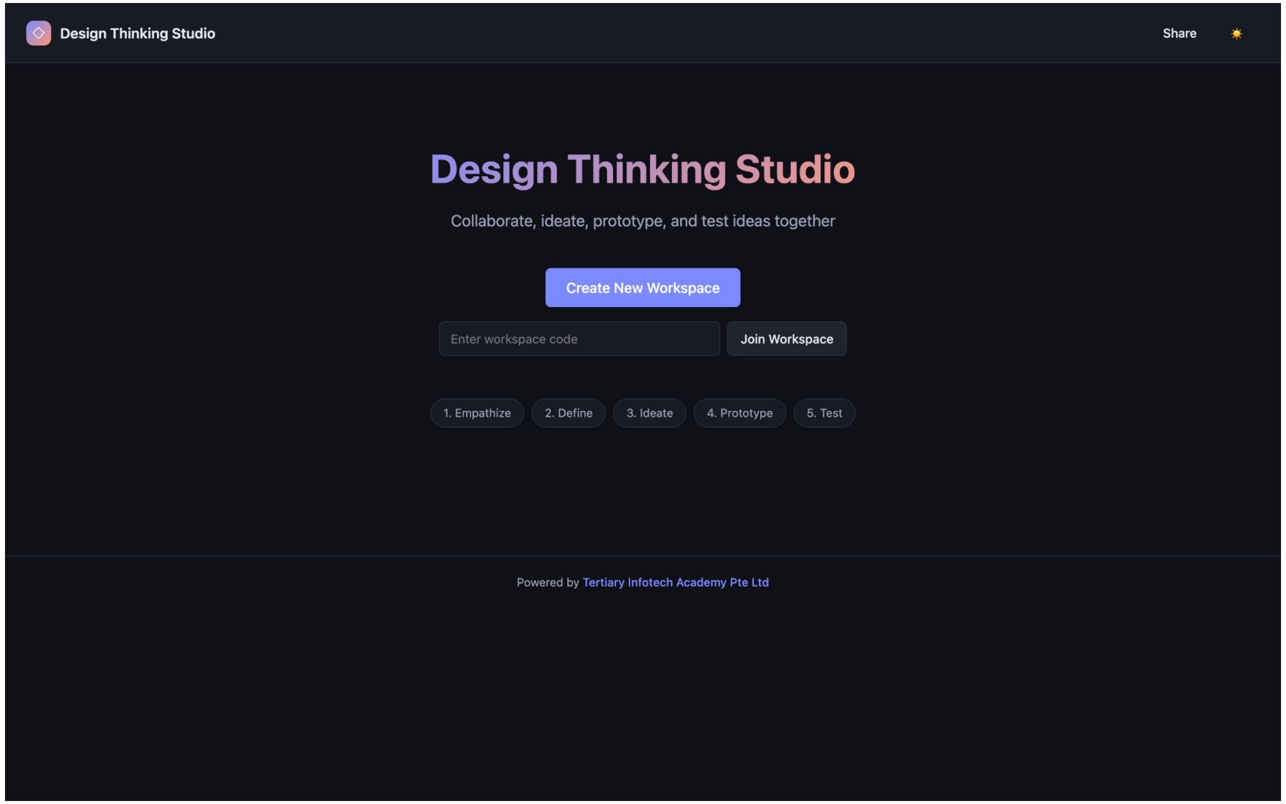
Tool: Fishbone
45 minutes · LO1

ALL ON ONE CASE

Every activity in this topic advances the same HarbourFront CustomerConnect case, so the artefacts you build compound.

ACTIVITY 1 · TOOL

The Tool — Design Thinking



Activity 1 · 45 minutes · full step-by-step: Learner Guide, Activity 1

Open it in your browser

https://alfredang.github.io/designthinking/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 1 · 45 MIN

Empathise with the Customer to Reframe a Failing Project

THE SITUATION

HarbourFront Logistics spent 11 months building the CustomerConnect portal from a signed-off 96-page requirements document. Six months after launch, only 3 of 24 features are used weekly. The top customer complaint to the call centre is still "where is my shipment right now?" — a question the portal was supposed to answer. You are the newly appointed Agile project lead. Before writing a single new requirement, you must understand the customer.

WHAT YOU'LL DO

Working in teams of 3–4 as the CustomerConnect project team, you use the Design Thinking tool to build an empathy map and a persona for Priya Menon, a warehouse operations executive at a HarbourFront customer who tracks 40–60 inbound shipments a week. You then reframe the original requirement statement into a customer-value problem statement and identify the three highest-value features to deliver first.

YOU'LL PRODUCE

A completed empathy map and persona for Priya Menon, a reframed problem statement, and a ranked list of the top 3 customer-value features with the reasoning for each.

DONE WHEN

Your reframed problem statement names a specific user, a specific need and a specific consequence — and contains no solution or technology. Your top 3 features each state a customer benefit. If any 'feature' is a module name or a format, it is still a requirement, not a value statement — rewrite it.

ACT 1

Tool: Design Thinking — https://alfredang.github.io/designthinking/

LO 1 · Analyse current and future customer needs and preferences using Design Thinking to reframe requirements around customer value (K2, A5). · Full step-by-step: Learner Guide, Activity 1

Debrief — Activity 1: What It Proves

1. The original project delivered exactly what was signed off, and still failed. Signed-off scope is not the same as delivered value.

2. Empathising took 45 minutes. It would have changed 11 months of build.

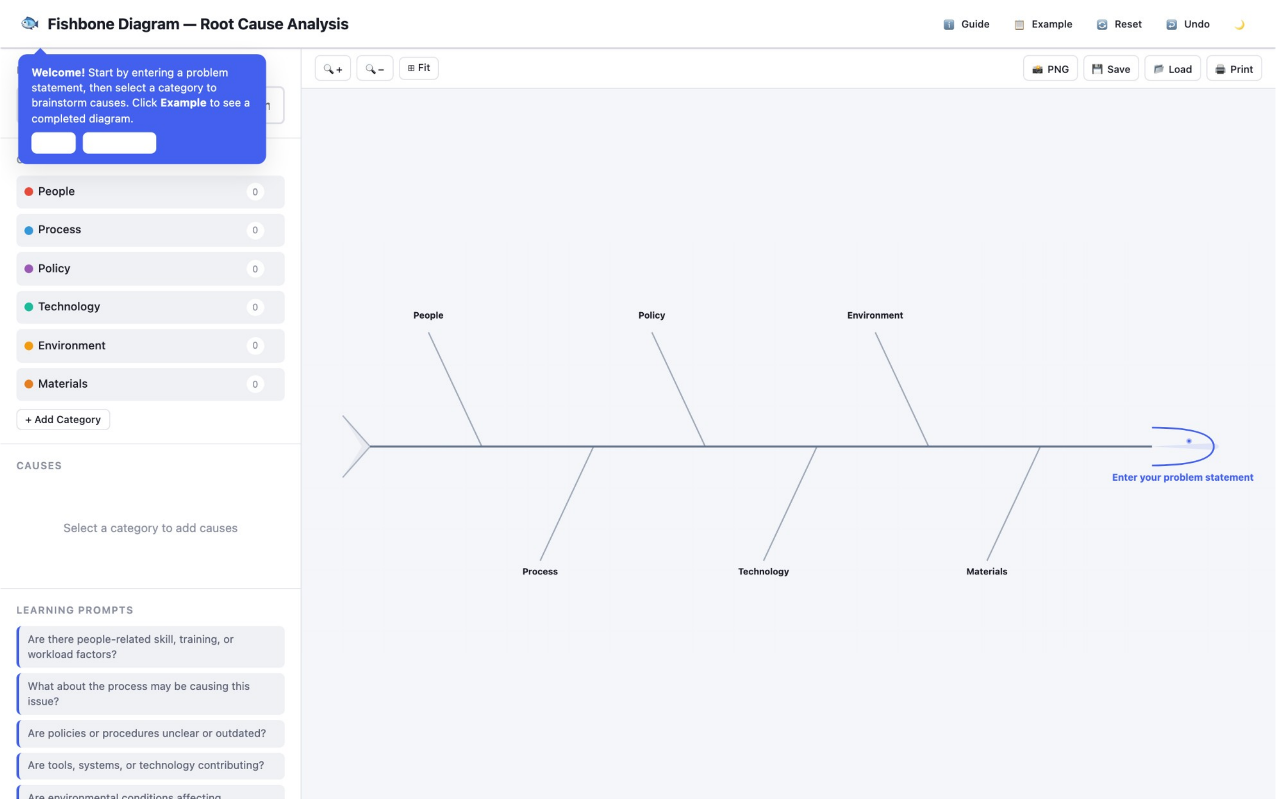
3. This is K2 in practice: analysing current and future customer needs is a repeatable method, not an intuition.

SUCCESS CRITERION

Your reframed problem statement names a specific user, a specific need and a specific consequence — and contains no solution or technology. Your top 3 features each state a customer benefit. If any 'feature' is a module name or a format, it is still a requirement, not a value statement — rewrite it.

ACTIVITY 2 · TOOL

The Tool — Fishbone



Activity 2 · 45 minutes · full step-by-step: Learner Guide, Activity 2

Open it in your browser

https://alfredang.github.io/fishbone/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 2 · 45 MIN

Diagnose the Waterfall Failure with a Fishbone Analysis

THE SITUATION

The HarbourFront management committee wants to know why CustomerConnect was 4 months late and 62% unused before it approves any new funding. The programme director's explanation was "the team underestimated". The CEO is not satisfied with that answer and has asked your team for a structured diagnosis by Friday.

WHAT YOU'LL DO

Your team uses the Fishbone (Ishikawa) tool to analyse the problem statement 'CustomerConnect delivered late and mostly unused'. You populate six cause categories, identify which causes are structural to the waterfall approach rather than failures of individual effort, and mark which ones an Agile approach would actually address. You then write a one-page recommendation to the committee.

YOU'LL PRODUCE

A completed fishbone diagram with 6 categories and at least 18 causes, each tagged as structural or behavioural, plus a one-page recommendation identifying which causes Agile addresses and which it does not.

DONE WHEN

Your diagram carries at least 18 causes across all 6 categories, every cause is tagged S or B, and each structural cause is paired with a named Agile practice. Your recommendation honestly identifies at least two causes Agile will not fix — a diagnosis that concludes 'Agile solves everything' has not been done properly.

ACT 2

Tool: Fishbone — https://alfredang.github.io/fishbone/

LO 1 · Analyse the current business operating landscape and diagnose why a waterfall delivery failed, using cause-and-effect analysis (K1, K3, A2). · Full step-by-step: Learner Guide, Activity 2

Debrief — Activity 2: What It Proves

1. Most causes tag as structural. The team did not fail; the approach concentrated all risk at the end.

2. This is why the cost-of-change curve matters: every one of these causes became expensive because it was discovered late.

3. K3 in practice: organisational policies and governance had to change too. Adopting Agile inside a stage-gate funding model produces theatre, not agility.

SUCCESS CRITERION

Your diagram carries at least 18 causes across all 6 categories, every cause is tagged S or B, and each structural cause is paired with a named Agile practice. Your recommendation honestly identifies at least two causes Agile will not fix — a diagnosis that concludes 'Agile solves everything' has not been done properly.

Recap — Introduction to Agile Project Management

1

Activity 1 · Empathise with the Customer to Reframe a Failing Project

Analyse current and future customer needs and preferences using Design Thinking to reframe requirements around customer value (K2, A5).

2

Activity 2 · Diagnose the Waterfall Failure with a Fishbone Analysis

Analyse the current business operating landscape and diagnose why a waterfall delivery failed, using cause-and-effect analysis (K1, K3, A2).

Lunch Break

1 hour · Digital attendance (PM) when you return

02

TOPIC 02

Agile Essentials

Manifesto values & principles · Scrum · Lean · Kanban · XP · Overcoming resistance · LO2 · K4, K6, K7, A1, A3

TOPIC 02 · THE FOUNDATION

Four values. Twelve principles. Everything else is machinery.

The Agile Manifesto was written in 2001 by 17 practitioners. It is one page long, and it is the reference every framework in this topic is derived from.

The Agile Manifesto — Four Values

"We value the left over the right" — both matter; the left matters MORE when they conflict

Individuals and interactions	**Working software**	**Customer collaboration**	**Responding to change**
OVER	OVER	OVER	OVER
processes and tools	comprehensive documentation	contract negotiation	following a plan

agilemanifesto.org — 'while there is value in the items on the right, we value the items on the left more'.

Reading the Four Values Correctly

1 Individuals & interactions

Tools are necessary; they never rescue a team that will not talk. Problems are solved by people.

2 Working software

Deliver something that works. Keep documents barely sufficient, just in time, and for a stated reason.

3 Customer collaboration

Manage change rather than suppress it. Build a shared, written definition of 'done'.

4 Responding to change

Energy spent defending the original plan is energy taken from delivering value.

THE COMMON MISREADING

'We value the left over the right' does NOT mean the right is worthless. Agile teams still plan, still document, still contract — they just refuse to let those things outrank delivered value.

The Twelve Principles — 1 to 6

1

1 · Satisfy the customer

Through early and continuous delivery of valuable work.

2

2 · Welcome change

Even late in development — harness change for competitive advantage.

3

3 · Deliver frequently

From a couple of weeks to a couple of months; prefer the shorter timescale.

4

4 · Work together daily

Business people and delivery people, throughout the project.

5

5 · Motivated individuals

Give them the environment and support they need, and trust them.

6

6 · Face-to-face conversation

The most efficient and effective way to convey information.

The Twelve Principles — 7 to 12

1

7 · Working output measures progress

Not percent-complete of tasks, and not documents produced.

2

8 · Sustainable pace

Sponsors, developers and users maintain a constant pace indefinitely.

3

9 · Technical excellence

Continuous attention to quality and good design enhances agility.

4

10 · Simplicity

Maximising the work NOT done is essential.

5

11 · Self-organising teams

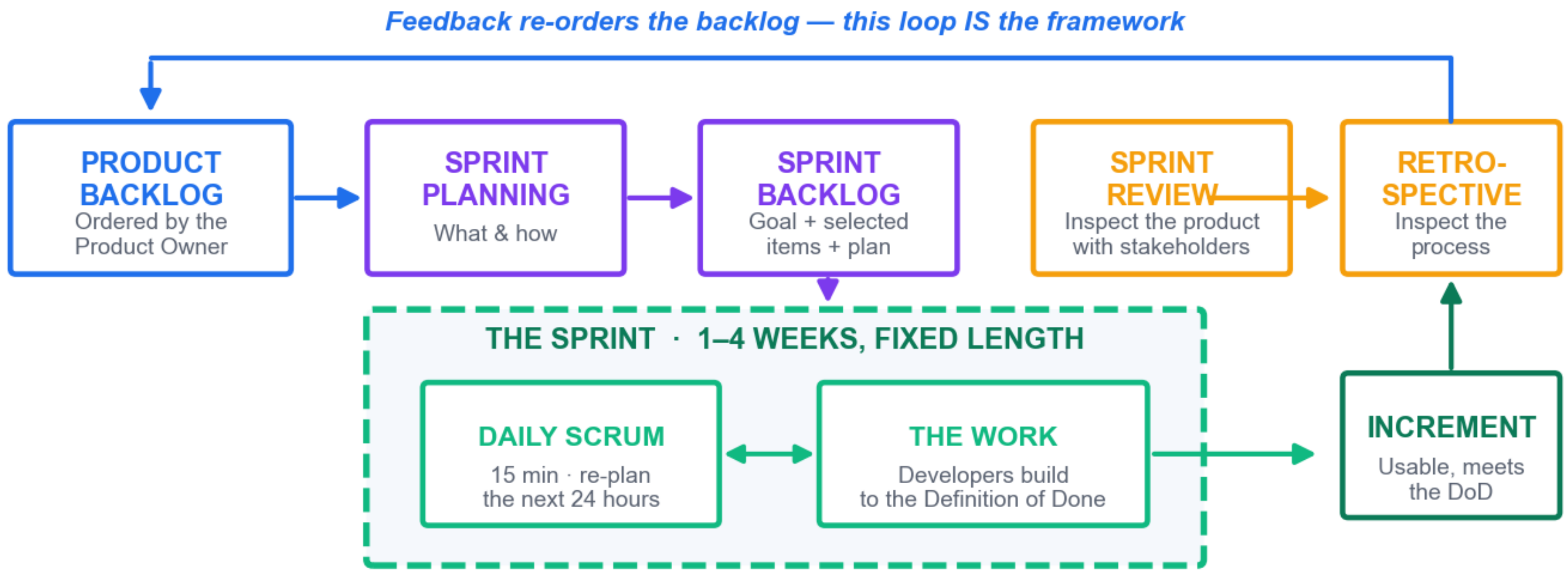
The best architectures, requirements and designs emerge from them.

6

12 · Reflect and adjust

At regular intervals, tune and adjust behaviour accordingly.

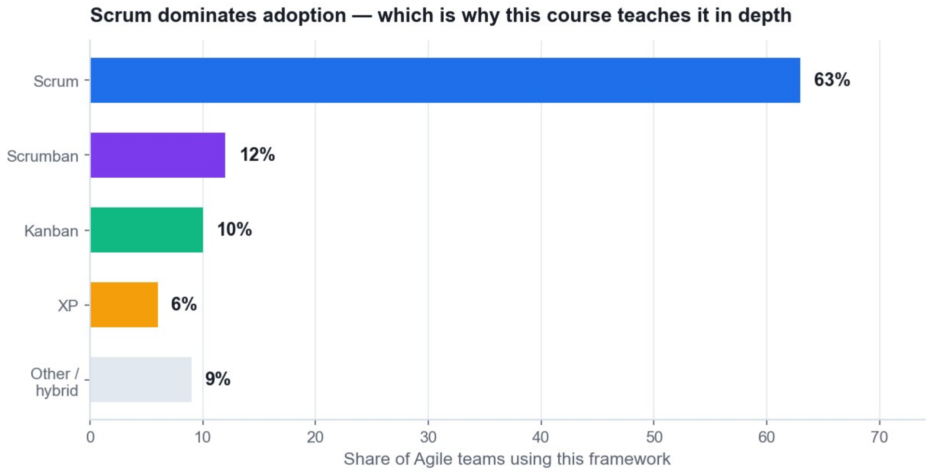
The Scrum Framework



3 accountabilities: Product Owner · Scrum Master · Developers   3 artefacts: Product Backlog · Sprint Backlog · Increment   5 events: Sprint · Planning · Daily Scrum · Review · Retrospective

Scrum is built on empiricism: transparency, inspection and adaptation. The loop is the framework.

Why We Teach Scrum in Depth



Indicative industry distribution (Coursera / State of Agile reporting). Framework choice should follow the work, not the fashion.

Scrum dominates adoption

Roughly 63% of Agile teams use Scrum, so it is the shared vocabulary.

It is prescriptive enough to learn

Fixed roles, artefacts and events give a new team something to follow.

It generalises

The same cadence works for marketing, operations and product teams.

Then tailor it

Shu-Ha-Ri: follow the framework first, adapt once you understand why.

The Three Scrum Accountabilities

1 Product Owner

Owns value and the ORDER of the product backlog. One person, not a committee. Accepts or rejects each increment.

2 Scrum Master

Owns the process. A servant leader who removes impediments, coaches the team and shields it from interruption.

3 Developers

Own HOW the work is done and how much enters the sprint. Cross-functional, self-organising, collectively accountable.

THE FAILURE MODE

Almost every 'Scrum doesn't work here' story traces to a Product Owner who cannot decide, or a manager who overrules the sprint. You will map this yourself in Activity 4.

Scrum Artefacts and Their Commitments

Artefact	What it is	Its commitment
Product Backlog	The ordered list of everything known to be needed	The Product Goal
Sprint Backlog	The selected items plus the plan to deliver them	The Sprint Goal
Increment	A usable, integrated slice of the product	The Definition of Done

Each artefact carries a commitment. An artefact without its commitment becomes a list nobody trusts.

The Five Scrum Events

1	2	3	4	5
Sprint Planning	**Daily Scrum**	**The Work**	**Sprint Review**	**Retrospective**
what and how · up to 8h	15 min · re-plan the next 24h	build to the Definition of Done	inspect the product with stakeholders	inspect the process

All five sit inside the Sprint — a fixed-length container of 1–4 weeks. The length does not change sprint to sprint.

Definition of Done

DEFINITION OF DONE — the team's shared, visible quality bar

- ✓ Code written and peer reviewed
- ✓ Unit and integration tests pass
- ✓ Acceptance criteria demonstrated
- ✓ Data-freshness check applied
  ← added in Activity 6 after the 5 Whys found the root cause
- ✓ Deployed to staging
- ✓ Accepted by the Product Owner

"Done" is not "nearly done". Work that fails the DoD returns to the backlog at full estimate — it is never counted as velocity.

The DoD is the team's shared quality bar. Without it, 'done' means something different to every person in the room.

Lean — Seven Principles from Toyota

1 **Eliminate waste**
To maximise value, minimise everything that is not value.

2 **Amplify learning**
Communicate early and often; get feedback as soon as possible.

3 **Decide as late as possible**
Keep options open until the last responsible moment.

4 **Deliver as fast as possible**
Speed and quality are allies, not opposites.

5 **Empower the team**
Respect the team's superior knowledge of the technical steps.

6 **Build quality in**
Assure quality throughout, rather than inspecting it at the end.

7 **Optimise the whole**
The system is more than the sum of its parts — beware local optimisation.

8 **Kaizen**
Small, frequent, team-owned improvement, forever.

The Eight Wastes

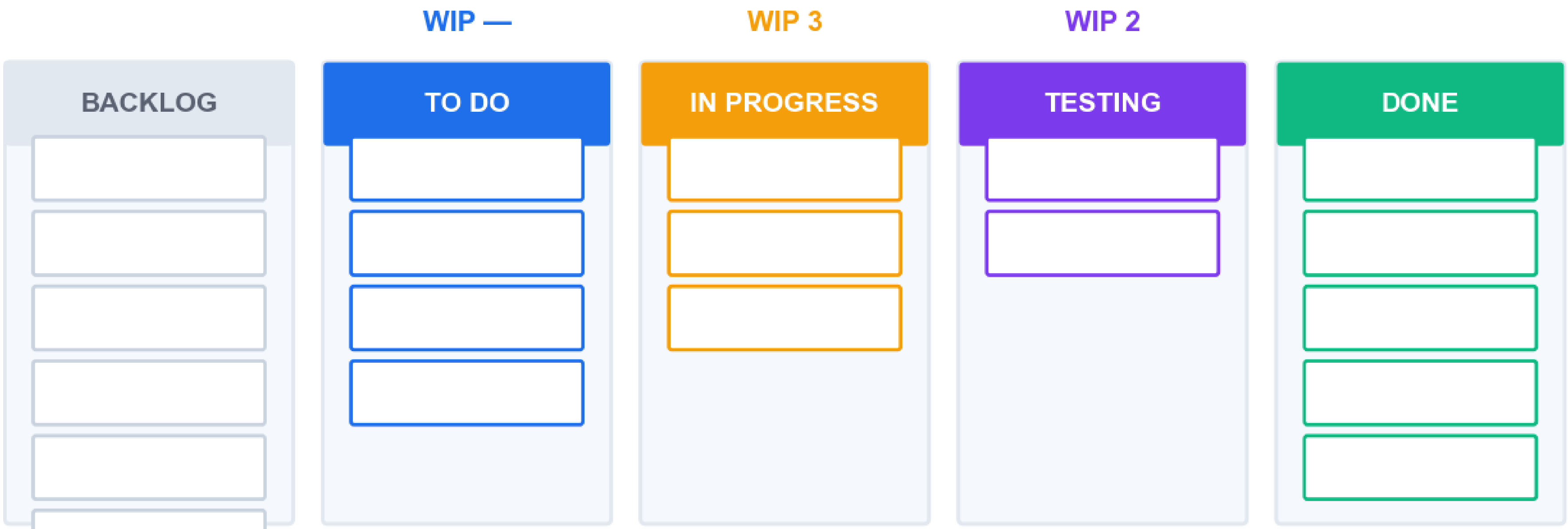
THE EIGHT WASTES — Lean asks which of these your process is paying for

Partially done work	**Extra processes**	**Extra features**	**Task switching**	**Waiting**	**Motion / handoffs**	**Defects**	**Unused talent**

In most knowledge work the largest waste is WAITING — and a value stream map is what exposes it

In knowledge work the dominant waste is almost always WAITING — which is why value stream mapping is so effective.

Kanban — Visualise, Limit WIP, Manage Flow



Little's Law: Cycle Time = Work in Progress ÷ Throughput → halve WIP at constant throughput and you halve cycle time

Little's Law is the engine: limiting WIP is what actually makes delivery faster, not working harder.

Kanban's Five Practices

1 Visualise the workflow

Knowledge work is invisible until you put it on a board.

2 Limit work in progress

WIP limits surface bottlenecks instead of hiding them.

3 Manage flow

Track how work moves, and measure the effect of each change.

4 Make policies explicit

Write down what 'ready' and 'done' mean so the team can debate them.

5 Improve collaborative ly

Evolve the process experimentally, as a team.

THE COUNTER-INTUITIVE PART

Starting less work finishes more work. You will prove this to yourself in Activity 5 when the burndown flattens, and again in Activity 8 with the control chart.

Scrum vs Kanban vs Scrumban — Choosing

Dimension	Scrum	Kanban	Scrumban
Cadence	Fixed sprints, 1–4 weeks	Continuous flow	Cadence plus WIP limits
Commitment	A Sprint Goal per sprint	Pull when capacity frees	Goal, with flexible pull
Best for	Teams of 5–9 with a product goal	Support queues, high variability	Teams outgrowing strict Scrum
Roles	PO, SM, Developers	No prescribed roles	Usually keeps PO and SM
Key metric	Velocity, sprint burndown	Cycle time, throughput, CFD	Both sets

Pick by the shape of the work, never by fashion. A support team forced into sprints will fight the framework every week.

Extreme Programming (XP)

1. **Five values**
   Simplicity, communication, feedback, courage and respect.

2. **Test-driven development**
   Write the test first; the code passes it once written correctly.

3. **Pair programming**
   Two people, real-time review, and knowledge spread through the team.

4. **Continuous integration**
   Bring code together constantly so incompatibility surfaces early.

5. **Refactoring**
   Remove redundancy and rejuvenate design continuously, not 'later'.

6. **Collective ownership**
   Anyone can improve any part; less risk when someone leaves.

7. **Small releases**
   Frequent small releases keep progress visible to the customer.

8. **Sustainable pace**
   Repeated long hours are unsustainable and counterproductive.

Other Methods Worth Knowing

1 DSDM / AgilePM

The APM-endorsed framework: feasibility, foundations, evolutionary development, deployment. Strong on governance.

2 Feature-Driven Development

Model first, build a feature list, then plan and build by feature.

3 Crystal

A family of methods sized by team and criticality, colour-coded.

4 SAFe / LeSS

Scaling frameworks for many teams on one product or portfolio.

FOR SINGAPORE PRACTITIONERS

DSDM/AgilePM is the framework APM (UK) endorses and the one most often referenced in regulated and public-sector delivery — worth knowing by name in a tender conversation.

Finding Agile Support — Address Each Fear Directly (K4)

1. **Senior management & sponsors**
   Fear: the risk of visible failure. Answer: incremental funding lets you stop early and cheaply.

2. **Middle managers**
   Fear: loss of control. Answer: they gain real progress data instead of percent-complete estimates.

3. **The project team**
   Fear: exposure and surveillance. Answer: the board shows the WORK, not individual performance.

4. **Users & customers**
   Fear: losing promised features. Answer: they get the highest-value features sooner, and a say each sprint.

5. **Finance & PMO**
   Fear: no fixed scope to audit. Answer: fixed cadence, fixed cost, auditable increments.

6. **Quality & compliance**
   Fear: reduced rigour. Answer: the Definition of Done makes the quality bar explicit and testable.

Change Management — Making the Adoption Stick (K4)

1 Start with a willing pilot

Volunteers, on a real project that matters, with visible sponsorship.

2 Make the wins visible

Publish the increment and the metrics. Evidence beats evangelism.

3 Evolutionary, not revolutionary

APM's guidance: change the culture in steps the organisation can absorb.

4 Train, coach, then withdraw

Coach the team through 2–3 sprints, then let it own the process.

THE HONEST WARNING

Never sell Agile as a way to get more output for less money. It buys responsiveness, earlier value and lower risk of building the wrong thing — not free capacity.

WHAT YOU'LL DO

Hands-On Activities — Agile Essentials

1 Activity 3 · Build the Product Backlog and Run Sprint 1 Planning

Tool: Scrum Board
60 minutes · LO2

2 Activity 4 · Clarify Agile Role Accountability with a RACI Matrix

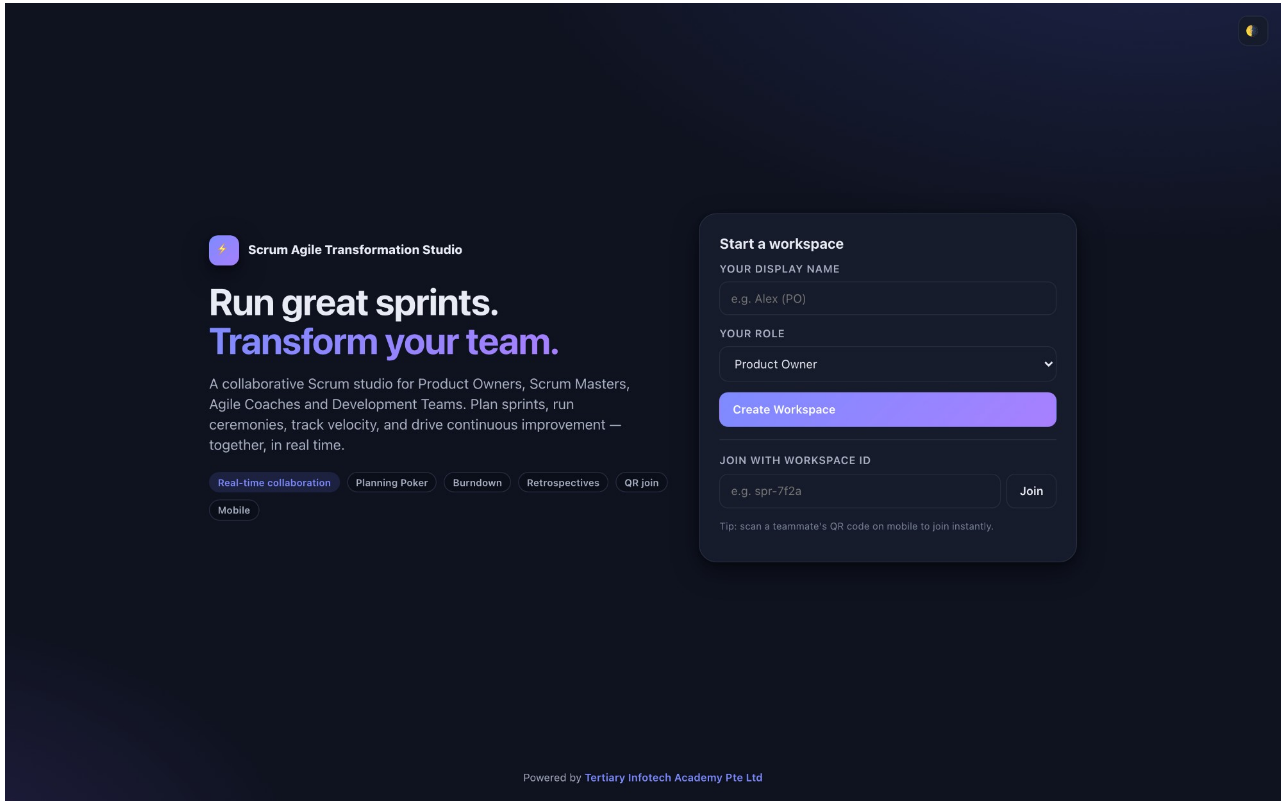
Tool: RACI Matrix
45 minutes · LO2

ALL ON ONE CASE

Every activity in this topic advances the same HarbourFront CustomerConnect case, so the artefacts you build compound.

ACTIVITY 3 · TOOL

The Tool — Scrum Board



Activity 3 · 60 minutes · full step-by-step: Learner Guide, Activity 3

Open it in your browser

https://alfredang.github.io/scrum/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 3 · 60 MIN

Build the Product Backlog and Run Sprint 1 Planning

THE SITUATION

HarbourFront's committee has approved a restart: 6 sprints of 2 weeks, one team of 7 people, and a hard demonstration to the top 5 customers in 12 weeks. Nothing from the old 96-page document is carried over unexamined. You are the team, with your trainer as Product Owner. Sprint 1 planning starts now.

WHAT YOU'LL DO

Your team writes user stories from the Activity 1 customer insight, applies INVEST, estimates with planning poker using the Fibonacci sequence, orders the backlog with MoSCoW, then loads Sprint 1 to a capacity of 20 story points and writes a single sprint goal. You use the Scrum tool to hold the backlog and the sprint board.

YOU'LL PRODUCE

A product backlog of at least 12 estimated user stories in priority order, a Sprint 1 backlog of about 20 points, one written sprint goal, and a Definition of Done agreed by the whole team.

DONE WHEN

Sprint 1 holds about 20 points, every story in it serves the one written sprint goal, no story exceeds 8 points, every story has Given/When/Then acceptance criteria, and the Definition of Done is visible to the whole team. A sprint whose stories serve four unrelated goals is a task list, not a sprint.

ACT 3

Tool: Scrum Board — https://alfredang.github.io/scrum/

LO 2 · Implement Agile practices by converting business needs into an ordered product backlog and a committed sprint backlog with a sprint goal (K6, K7, A3). · Full step-by-step: Learner Guide, Activity 3

Debrief — Activity 3: What It Proves

1. The sprint goal is the single most skipped artefact and the one that makes the sprint reviewable.

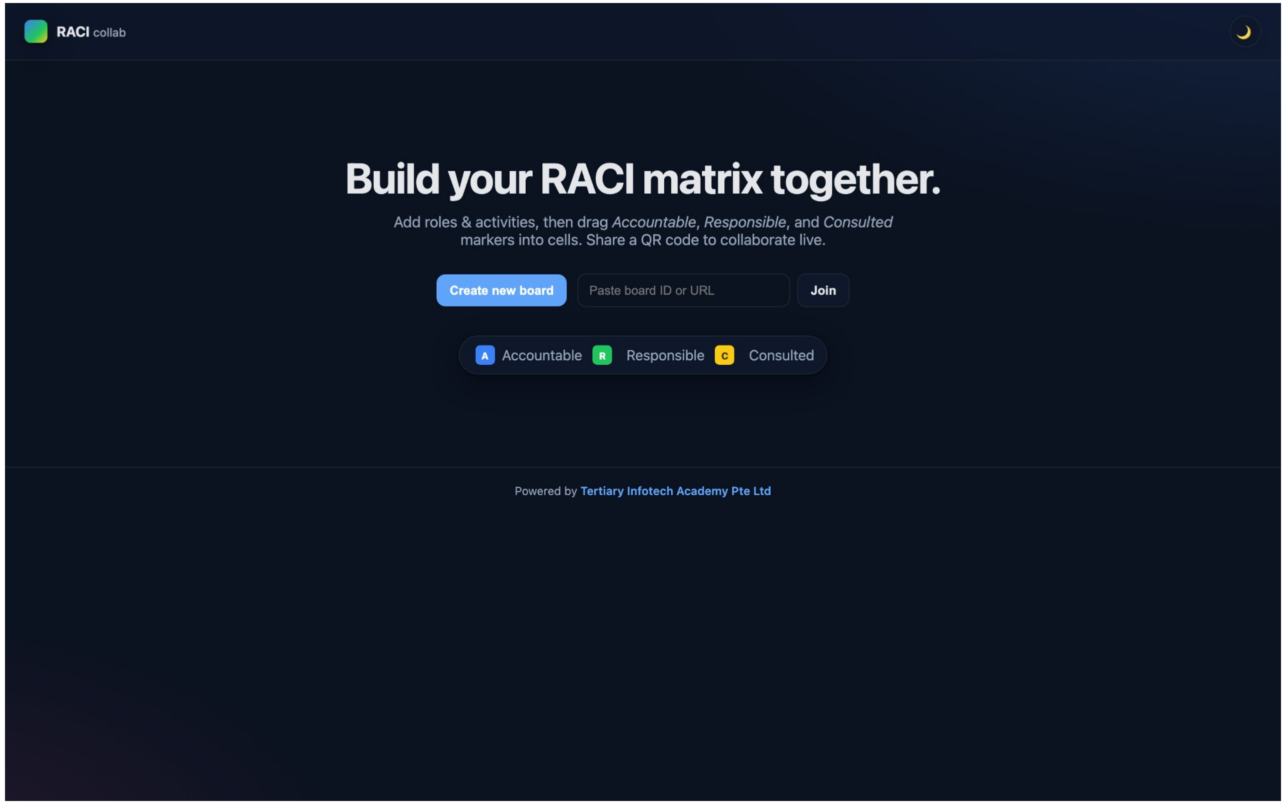
2. Splitting by workflow step keeps every slice demonstrable; splitting by technical layer produces a sprint with nothing to show.

3. This is A3 in practice: implementing Agile practices to reduce waste — an ordered backlog stops the team building the 62% nobody used.

SUCCESS CRITERION

Sprint 1 holds about 20 points, every story in it serves the one written sprint goal, no story exceeds 8 points, every story has Given/When/Then acceptance criteria, and the Definition of Done is visible to the whole team. A sprint whose stories serve four unrelated goals is a task list, not a sprint.

The Tool — RACI Matrix



Activity 4 · 45 minutes · full step-by-step: Learner Guide, Activity 4

Open it in your browser

https://alfredang.github.io/raci/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 4 · 45 MIN

Clarify Agile Role Accountability with a RACI Matrix

THE SITUATION

Two weeks into the restart, CustomerConnect has stalled in a familiar way. The old programme director is still approving every story change. The Scrum Master has been asked to produce a weekly percent-complete report. Two developers have escalated that they receive conflicting priorities from the Product Owner and from the operations manager. Nobody is behaving badly — the accountabilities were never made explicit.

WHAT YOU'LL DO

Your team uses the RACI tool to map 12 real project decisions and activities against the five roles in play. You then find the anti-patterns — activities with two Accountables, activities with none, and roles consulted on everything — and produce the corrected matrix that resolves the conflicting-priorities escalation.

YOU'LL PRODUCE

A completed RACI matrix covering 12 activities across 5 roles, a list of the anti-patterns found, and the corrected matrix with the specific change that resolves the developers' escalation.

DONE WHEN

Every one of the 12 rows has exactly one A. No role is C on more than about half the rows. The developers' conflicting-priorities escalation is traceable to a specific duplicate A in your first draft, and your corrected matrix removes it.

ACT 4

Tool: RACI Matrix — https://alfredang.github.io/raci/

LO 2 · Share information across teams and organise work in alignment with priorities by making Agile role accountability explicit (K5, A1, A2). · Full step-by-step: Learner Guide, Activity 4

Debrief — Activity 4: What It Proves

1. Most Agile 'transformation failures' are accountability failures wearing an Agile label.

2. The single A rule is what makes a Product Owner viable. Two Accountables for backlog order guarantees the escalation you just resolved.

3. This is A1 and A2 in practice: sharing information across teams to bridge operational barriers, and organising work in line with actual priorities.

SUCCESS CRITERION

Every one of the 12 rows has exactly one A. No role is C on more than about half the rows. The developers' conflicting-priorities escalation is traceable to a specific duplicate A in your first draft, and your corrected matrix removes it.

TOPIC RECAP · LO2 · K4, K6, K7, A1, A3

Recap — Agile Essentials

1

Activity 3 · Build the Product Backlog and Run Sprint 1 Planning

Implement Agile practices by converting business needs into an ordered product backlog and a committed sprint backlog with a sprint goal (K6, K7, A3).

2

Activity 4 · Clarify Agile Role Accountability with a RACI Matrix

Share information across teams and organise work in alignment with priorities by making Agile role accountability explicit (K5, A1, A2).

End of Day 1

See you at 09:30 tomorrow · Digital attendance (AM)

03

TOPIC 03

Agile Project Execution and Tracking

Team building & vision · User stories & estimation · Execution · Metrics · Continuous improvement · LO3 · K5, A4, A6, A7

TOPIC 03 · MAKING IT RUN

A team, a vision, a rhythm, and honest numbers.

Topic 3 is where Agile stops being a philosophy and becomes a week of work: who is on the team, what they are aiming at, how the work flows, and what the metrics actually tell you.

Team Composition and Formation Models (K5)

1 Cross-functional

Every skill needed to get to Done sits inside the team.

2 Self-organising

The team decides how to do the work, not just what to do.

3 Fewer than ~12

Communication paths grow faster than headcount. Small teams stay coherent.

4 Stable membership

Velocity is meaningless if the team changes every sprint.

5 Generalising specialists

Deep in one skill, capable across several — this removes queues.

6 Co-located or deliberately connected

Osmotic communication, or tooling that replaces it on purpose.

THE BOTTLENECK YOU WILL MEET

One specialist per skill guarantees queues — work piles up behind the only person who can do it. You will see exactly this as the Testing bottleneck in Activity 8.

Tuckman's Stages and Adaptive Leadership

ADAPTIVE LEADERSHIP — match your style to the team's stage, not your preference

FORMING	STORMING	NORMING	PERFORMING	ADJOURNING
Polite, unsure Low output	Conflict surfaces Output dips	Norms agreed Output climbs	Self-organising High output	Work closes Team disbands
DIRECTING	COACHING	SUPPORTING	DELEGATING	RECOGNISING

The storming dip is normal and necessary. A team that never storms is usually avoiding, not aligned.

Match your leadership style to the team's stage. Delegating to a forming team fails; directing a performing team insults it.

Servant Leadership — What the Agile Leader Actually Does

1 **Shield the team**
Absorb interruptions so the team can hold a sustainable focus.

2 **Remove impediments**
Own the blockers the team cannot clear itself, and clear them fast.

3 **Re-communicate the vision**
People forget the goal under delivery pressure. Repeat it.

4 **Tap intrinsic motivation**
Autonomy, mastery and purpose outperform instruction.

5 **Model the behaviour**
Honesty, competence and forward-looking judgement, visibly.

6 **Create safety to experiment**
A team that cannot fail safely will not tell you the truth.

Setting the Shared Vision

1. **Agile charter**
   Who, what, why, when, where and how — high-level, and authority to proceed.

2. **Product vision statement**
   One paragraph any team member can repeat from memory.

3. **Definition of Done**
   The shared, testable meaning of 'finished'.

4. **Personas**
   Grounded, goal-oriented descriptions of the real users.

5. **Wireframes & prototypes**
   Low-fidelity artefacts that make 'done' visible before it is built.

6. **Journey and story maps**
   The whole experience, so the team can see what each slice contributes.

The Anatomy of a User Story

As a **warehouse operations executive,**
I want **a live customs status on each inbound shipment,**
so that **I can re-sequence today's production.**

WHO — the role
WHAT — the goal
WHY — the benefit

ACCEPTANCE CRITERIA

GIVEN a shipment in customs clearance
WHEN I open the shipment card
THEN I see the current state and its last-updated time

INVEST

Independent · Negotiable · Valuable
Estimatable · Small · Testable

Fails 'Small'? Split by workflow step — never by technical layer.

Role, goal, benefit — plus testable acceptance criteria. This exact story is the one you write in Activity 3.

The Three Cs and INVEST

1 Card

The story is a placeholder, deliberately too small to hold the full requirement.

2 Conversation

The detail is developed by talking, not by writing a longer document.

3 Confirmation

Acceptance criteria state how you will know it is done.

4 INVEST

Independent · Negotiable · Valuable · Estimatable · Small · Testable.

THE SPLIT RULE

A story too large for one sprint must be split by WORKFLOW STEP or by DATA TYPE — never by technical layer. Splitting into 'the database part' and 'the UI part' produces a sprint with nothing to demonstrate.

Prioritisation Techniques

1. **MoSCoW**
   Must / Should / Could / Won't have this time. Used in Activity 3.

2. **Dot voting**
   Each person distributes a fixed number of dots across the options.

3. **100-point method**
   Each stakeholder allocates 100 points, forcing real trade-offs.

4. **Monopoly money**
   Equal 'funds' spent on what each stakeholder values most.

5. **Kano analysis**
   Separates delighters, satisfiers, must-haves and indifferent features.

6. **Weighted shortest job first**
   Value divided by effort — highest ratio goes first.

Estimation — Why Relative Beats Absolute

1 Humans estimate badly in absolutes

We are poor at hours, and consistently good at comparisons.

2 Story points

Complexity, effort and risk in one relative number.

3 Planning poker

Simultaneous reveal on a Fibonacci scale defeats anchoring and the loudest voice.

4 Affinity & T-shirt sizing

Fast grouping for large backlogs before detailed estimation.

THE 13-POINT RULE

A story estimated at 13 or 21 points is not a big story — it is a story the team does not yet understand. Split it before it enters a sprint.

Timeboxing and Parkinson's Law

1. **SET THE STAGE** — 5 min
2. **GATHER DATA** — 40 min
3. **GENERATE INSIGHTS** — 25 min
4. **DECIDE WHAT TO DO** — 20 min
5. **CLOSE THE RETRO** — 20 min

≈ 2 hours for a 2-week sprint · Insight tools: 5 Whys · Fishbone · Pareto · dot voting

Daily Scrum · 15 minutes

Long enough to re-plan a day, short enough to stay standing.

Retrospective · ~2 hours

For a two-week sprint, in five deliberate stages.

Sprint · 1–4 weeks

Fixed length, so velocity means something across sprints.

Parkinson's Law

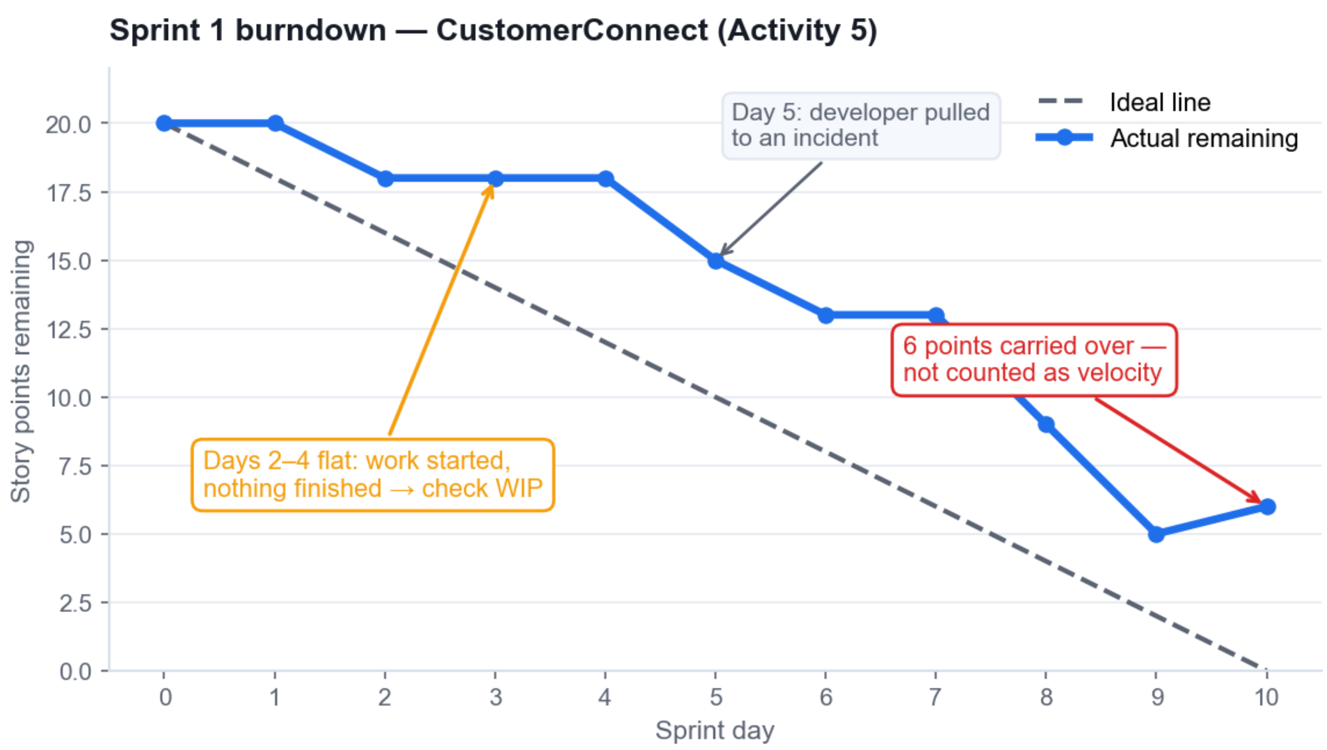
Work expands to fill the time available — so the box does the managing.

TOPIC 03 · A4 · MEASURING PROGRESS

Five metrics tell you everything you need to know.

Sprint burndown for the sprint. Release burndown for the release. Velocity for the forecast. Control chart for cycle time. Cumulative flow diagram for the bottleneck.

Metric 1 — Sprint Burndown



What it shows

Points remaining inside one sprint, day by day, against the ideal line.

Read the flat line

Flat means work is STARTED but not FINISHED — check WIP, not effort.

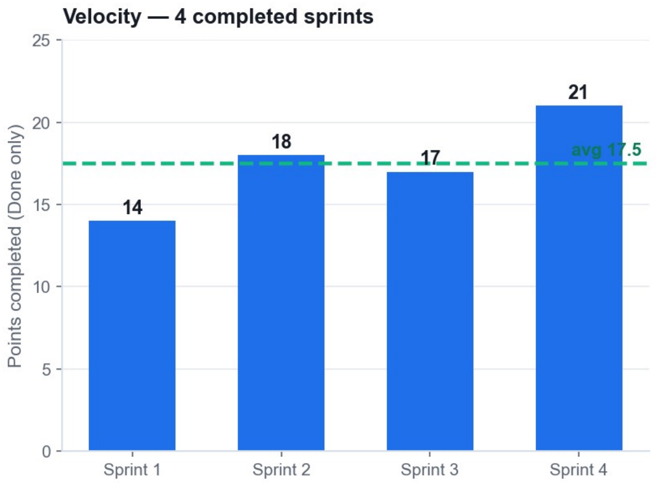
Read the cliff

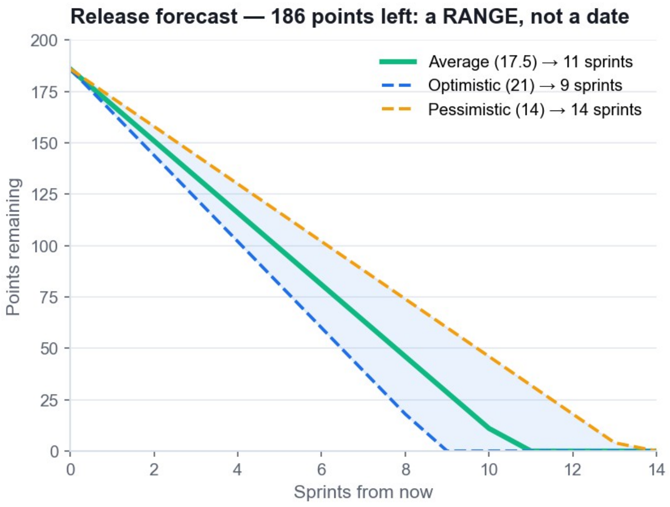
A late vertical drop means work was integrated at the end. That is a risk, not a triumph.

Carryover

Unfinished work returns to the backlog at full estimate. It is never partial velocity.

Metrics 2 & 3 — Velocity and the Release Forecast





Velocity

Average points DONE per sprint. A forecasting input, never a performance target.

Forecast as a range

Slowest, average and fastest sprints give a cone, not a date.

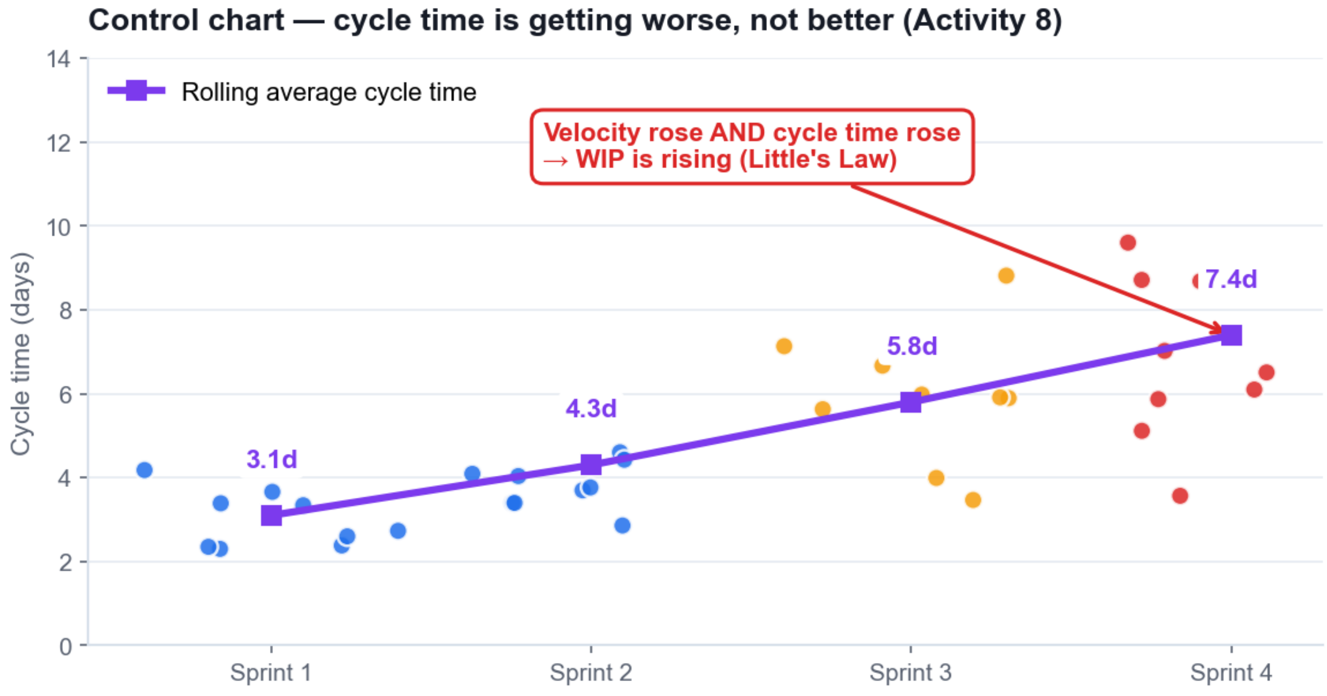
State the assumptions

Stable team, stable backlog, same estimation scale, no lost sprints.

The anti-pattern

Reward velocity and you get estimate inflation — the forecast then breaks.

Metric 4 — The Control Chart



What it shows

Cycle time and lead time per work item, and the trend across sprints.

Lead vs cycle time

Lead time is the customer's whole wait; cycle time is the team's active portion.

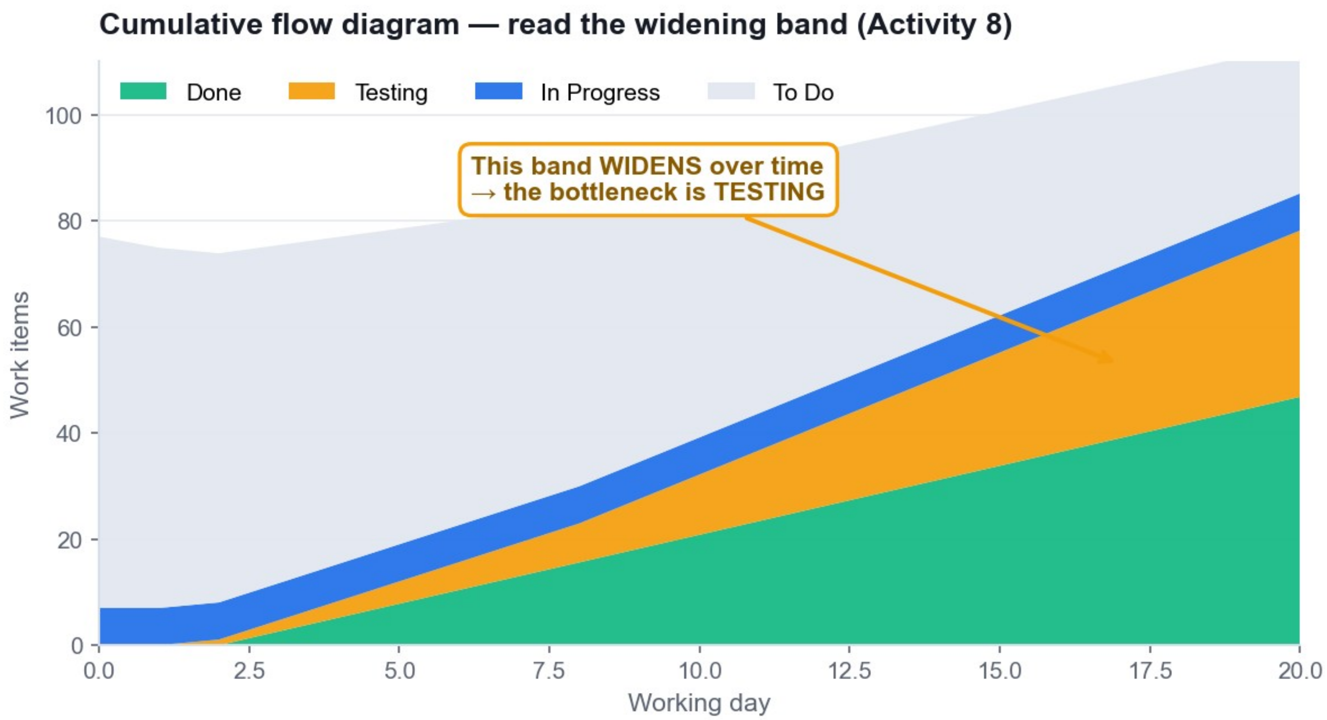
The warning sign

Velocity rising AND cycle time rising means WIP is rising.

What to do

Lower the WIP limit at the constraint. Stop starting, start finishing.

Metric 5 — The Cumulative Flow Diagram



What it shows

The count of items in each status over time, as stacked bands.

Read the widening band

A band widening vertically is a bottleneck at THAT status.

Read the flat Done band

Flat 'Done' means nothing is being completed, whatever the team is busy with.

Act on the constraint

Move capacity to the widening column — not to the busiest-looking one.

Little's Law — Why Limiting WIP Speeds Delivery

Scenario	WIP	Throughput	Cycle time
Team starts everything	30 items	5 items/week	6 weeks
Team halves WIP	15 items	5 items/week	3 weeks
Team quarters WIP	8 items	5 items/week	1.6 weeks

Throughput did not change in any row. Only WIP changed — and cycle time fell with it. This is arithmetic, not opinion.

The Retrospective in Five Stages

1 SET THE STAGE — 5 min

2 GATHER DATA — 40 min

3 GENERATE INSIGHTS — 25 min

4 DECIDE WHAT TO DO — 20 min

5 CLOSE THE RETRO — 20 min

≈ 2 hours for a 2-week sprint · Insight tools: 5 Whys · Fishbone · Pareto · dot voting

The retrospective is the engine of improvement. Without an owned, funded action, it is just a feelings meeting.

Root-Cause Tools — Use Them Together

1 5 Whys

Drills ONE causal chain to a process cause you can change. Activity 6.

2 Fishbone

Spreads causes across categories to see the whole problem. Activity 2.

3 Pareto

Shows which few causes carry most of the pain. Activity 7.

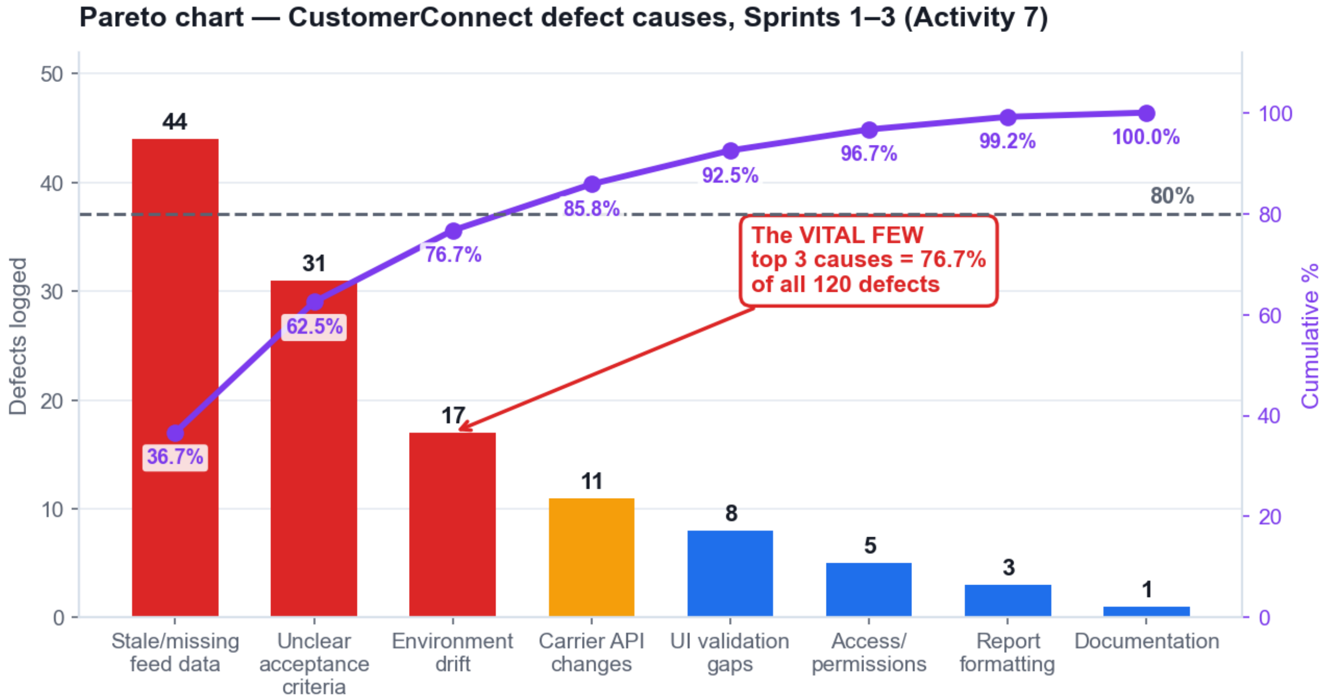
4 PDCA / Kaizen

Plan-Do-Check-Act: the loop that carries the fix into the next sprint.

THE SEQUENCE THAT WORKS

Fishbone to see the whole space, Pareto to pick the few that matter, 5 Whys to reach the changeable cause, then one SMART action with an owner and points in the next sprint.

Pareto — Which Few Causes Carry Most of the Pain



The real CustomerConnect defect data you will analyse in Activity 7 — 120 defects across 8 causes.

The 80/20 pattern

A small number of causes generate most of the defects — reliably, across domains.

Always sort descending

An unsorted Pareto chart cannot be read. The cumulative line depends on the order.

Find the crossing point

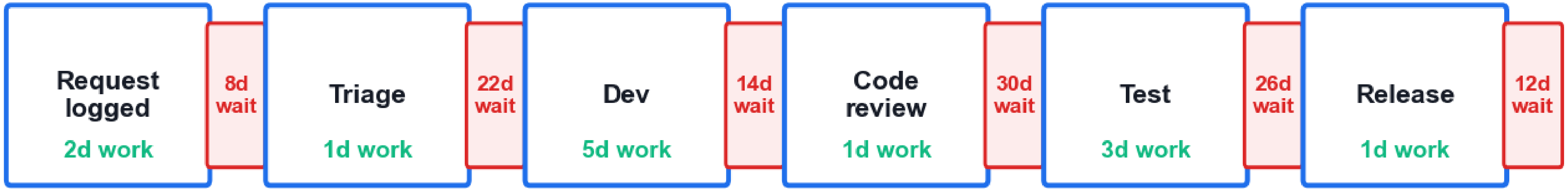
Read where the cumulative line crosses 80% — those are the vital few.

Decide, don't admire

The chart's purpose is to justify what you will NOT fix this sprint.

Value Stream Mapping — Where the Time Actually Goes

VALUE STREAM MAP — the waiting between steps dwarfs the work inside them



Processing time 13 days · Waiting time 112 days · Lead time 125 days · Efficiency 10%

Attack the red boxes first. Making the blue boxes faster changes almost nothing.

13 days of work inside 125 days of lead time — 10% efficiency. The opportunity is in the waiting, not the working.

Risk as Anti-Value

1. **Risk is negative value**
   It erodes or removes value, so it belongs in the same conversation as value.

2. **Risk-adjusted backlog**
   High-risk, high-value items are pulled earlier to retire uncertainty.

3. **Expected monetary value**
   Probability × impact, so risks can be compared honestly.

4. **Risk burndown chart**
   Tracks total exposure falling sprint by sprint.

5. **Spikes**
   A timeboxed investigation to reduce a specific technical or risk unknown.

6. **Schedule risk work as real items**
   A risk register nobody funds changes nothing.

Technical Debt and Ownership (A7)

1 Debt compounds

Work skipped to go faster makes every later sprint slower.

2 Fund refactoring inside the sprint

'Later' never arrives on its own.

3 Collective commitment

The team commits to the Sprint Goal together, then each member pulls their own work.

4 Raise impediments same-day

Surfacing a blocker on day 9 is the failure — not the blocker itself.

WHAT ACCOUNTABILITY LOOKS LIKE

An honest board, a same-day impediment, and a 'Done' that means Done. Every metric in this topic depends on those three behaviours.

WHAT YOU'LL DO

Hands-On Activities — Agile Project Execution and Tracking

1 Activity 5 · Execute Sprint 1 and Track It on the Scrum Board

Tool: Scrum Board
60 minutes · LO3

2 Activity 6 · Run the Sprint Retrospective with 5 Whys Root-Cause Analysis

Tool: 5 Whys
45 minutes · LO3

3 Activity 7 · Prioritise Defect Causes with a Pareto Chart

Tool: Pareto Chart
45 minutes · LO3

4 Activity 8 · Forecast the Release from Velocity and Read the Agile Metrics

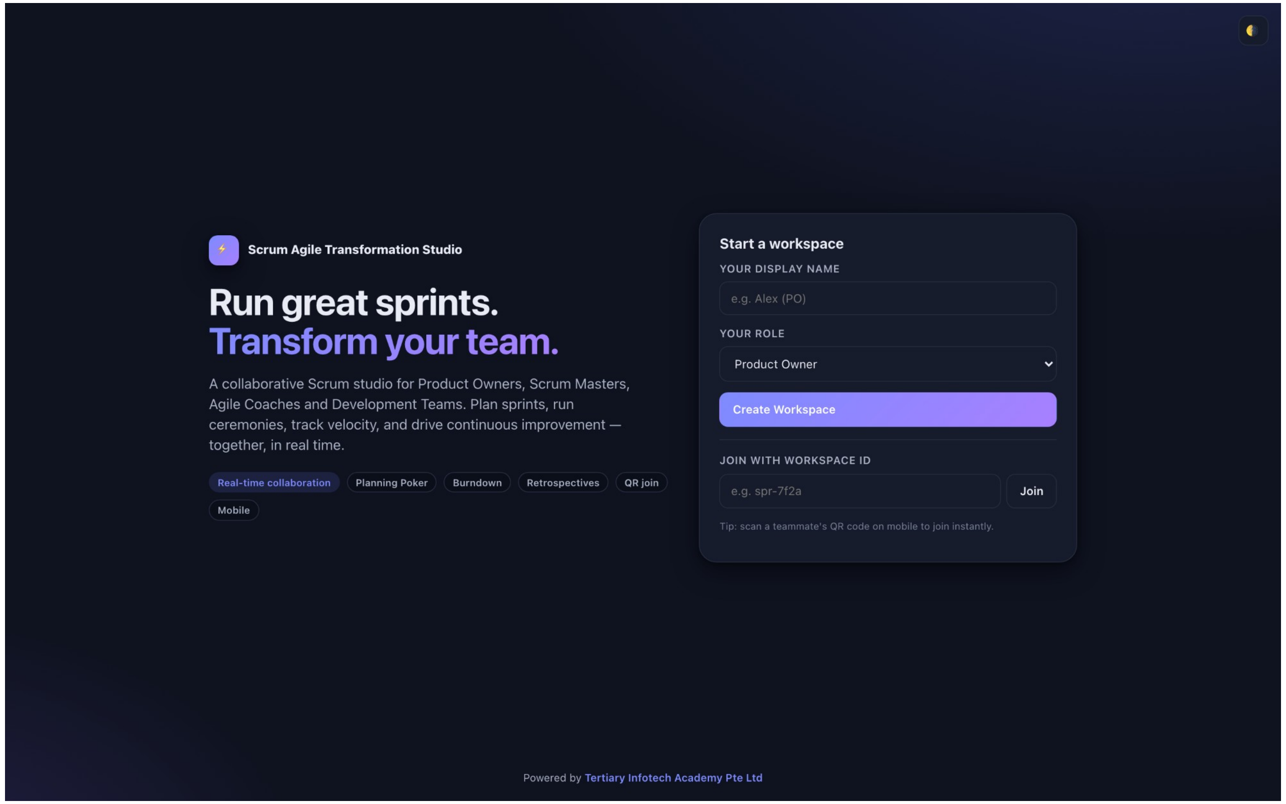
Tool: Scrum Board
45 minutes · LO3

ALL ON ONE CASE

Every activity in this topic advances the same HarbourFront CustomerConnect case, so the artefacts you build compound.

ACTIVITY 5 · TOOL

The Tool — Scrum Board



Activity 5 · 60 minutes · full step-by-step: Learner Guide, Activity 5

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Build the artefact

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Export it

Screenshot or export into your Activity worksheet.

TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 5 · 60 MIN

Execute Sprint 1 and Track It on the Scrum Board

THE SITUATION

Sprint 1 of the CustomerConnect restart runs for 10 working days with a committed 20 points. Your team will simulate all 10 days in compressed time, holding a daily stand-up each 'day', moving cards, and hitting three real impediments the trainer injects: the carrier data feed returns stale timestamps on Day 3, a developer is pulled onto a production incident on Day 5, and the Product Owner requests a new story mid-sprint on Day 7.

WHAT YOU'LL DO

Using your Sprint 1 backlog from Activity 3, your team runs the sprint day by day on the Scrum board. Each simulated day you hold a 3-minute stand-up answering the three questions, move cards across To Do / In Progress / Testing / Done, respect a WIP limit of 3 in progress, log impediments, and update the burndown. You handle each injected impediment as the framework prescribes.

YOU'LL PRODUCE

A completed 10-day sprint board, a burndown chart with all 10 data points, an impediment log with the resolution of each of the 3 injections, and the actual velocity achieved.

DONE WHEN

Your burndown has 10 plotted points and an ideal line. Your velocity counts only fully-Done stories, with carryover recorded separately. All 3 impediments are logged with a decision and a rationale. If your burndown is a straight diagonal line, you have plotted the plan, not the sprint — plot what actually happened.

ACT 5

Tool: Scrum Board — https://alfredang.github.io/scrum/

LO 3 · Measure progress against targets on a regular basis by executing a sprint on a live board and reading the burndown (K5, A4, A7). · Full step-by-step: Learner Guide, Activity 5

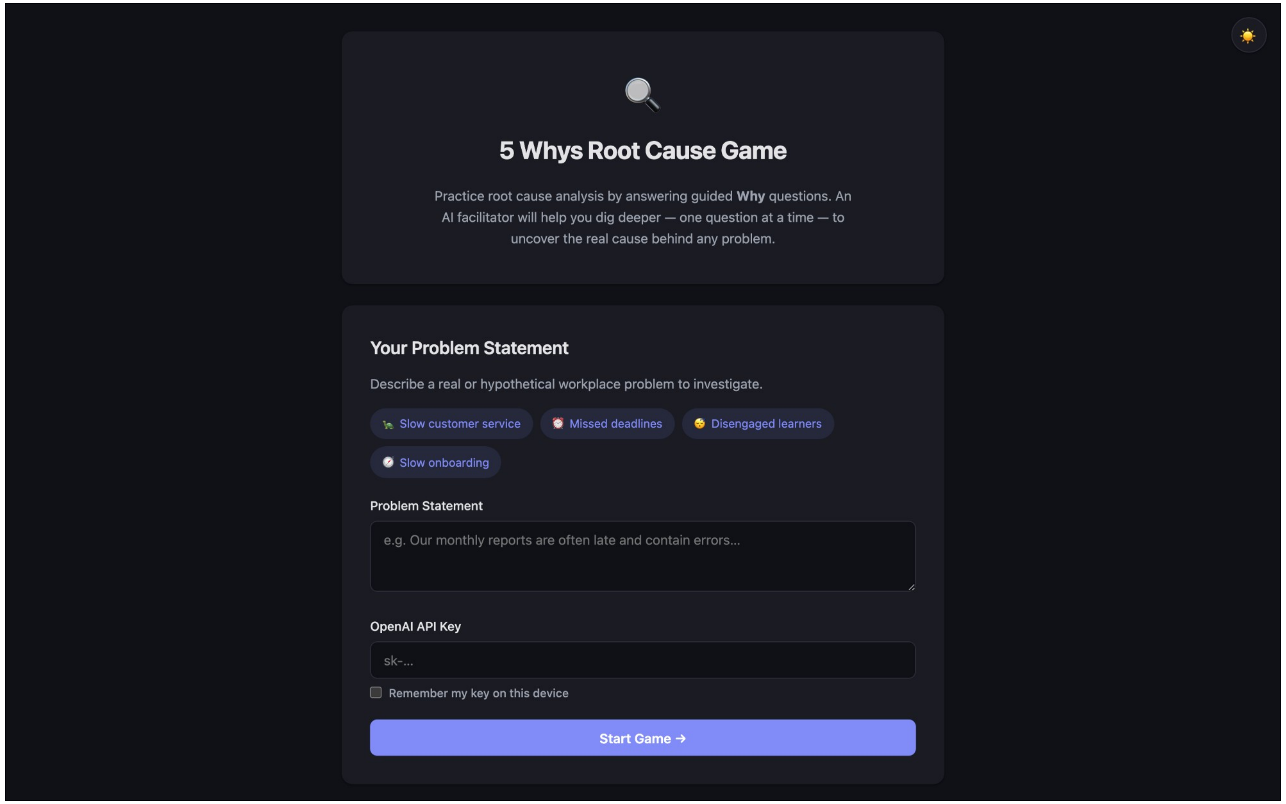
Debrief — Activity 5: What It Proves

1. The flat burndown on Day 3 is the most valuable thing on the chart. It is an early warning, and it only appears if the board is honest.
2. Carryover counted as velocity destroys forecasting. Velocity must mean 'Done', or every forecast built on it is fiction.
3. The Day 7 injection is the real test of Agile discipline: welcoming change does not mean accepting unbounded scope inside a committed sprint.

SUCCESS CRITERION

Your burndown has 10 plotted points and an ideal line. Your velocity counts only fully-Done stories, with carryover recorded separately. All 3 impediments are logged with a decision and a rationale. If your burndown is a straight diagonal line, you have plotted the plan, not the sprint — plot what actually happened.

The Tool — 5 Whys



Activity 6 · 45 minutes · full step-by-step: Learner Guide, Activity 6

Open it in your browser

https://alfredang.github.io/5whys/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 6 · 45 MIN

Run the Sprint Retrospective with 5 Whys Root-Cause Analysis

THE SITUATION

Sprint 1 finished 6 of 20 committed points short. In the review, one customer noted that the shipment status feature 'works but shows yesterday's data'. In the retrospective, the team's first instinct is to blame the carrier data feed. The Scrum Master pushes for a real root cause, because the same class of problem appeared twice in the old waterfall project.

WHAT YOU'LL DO

Your team runs a full retrospective in the five stages, using the 5 Whys tool on the single highest-impact problem from Sprint 1. You drive past the first plausible answer to a root cause the team can actually act on, then convert it into one SMART action with a named owner that enters the Sprint 2 backlog as a real item.

YOU'LL PRODUCE

A completed 5 Whys chain of at least 5 levels reaching an actionable root cause, plus one SMART improvement action with a named owner, a measure, and a place in the Sprint 2 backlog.

DONE WHEN

Your chain reaches a process or system cause the team can change, not a person and not an external party. Reading it backwards with 'because' holds together at every link. Your SMART action has an owner, a measure and points in the Sprint 2 backlog. If your root cause is 'the carrier's fault', you stopped at Why 1.

ACT 6

Tool: 5 Whys — https://alfredang.github.io/5whys/

LO 3 · Assess work performance and quality to ensure continuous improvement, drilling to the root cause of a delivery problem (A6, A7). · Full step-by-step: Learner Guide, Activity 6

Debrief — Activity 6: What It Proves

1. Stopping at 'the carrier feed was stale' would have produced a ticket. Reaching the copied Definition of Done produced a systemic fix.

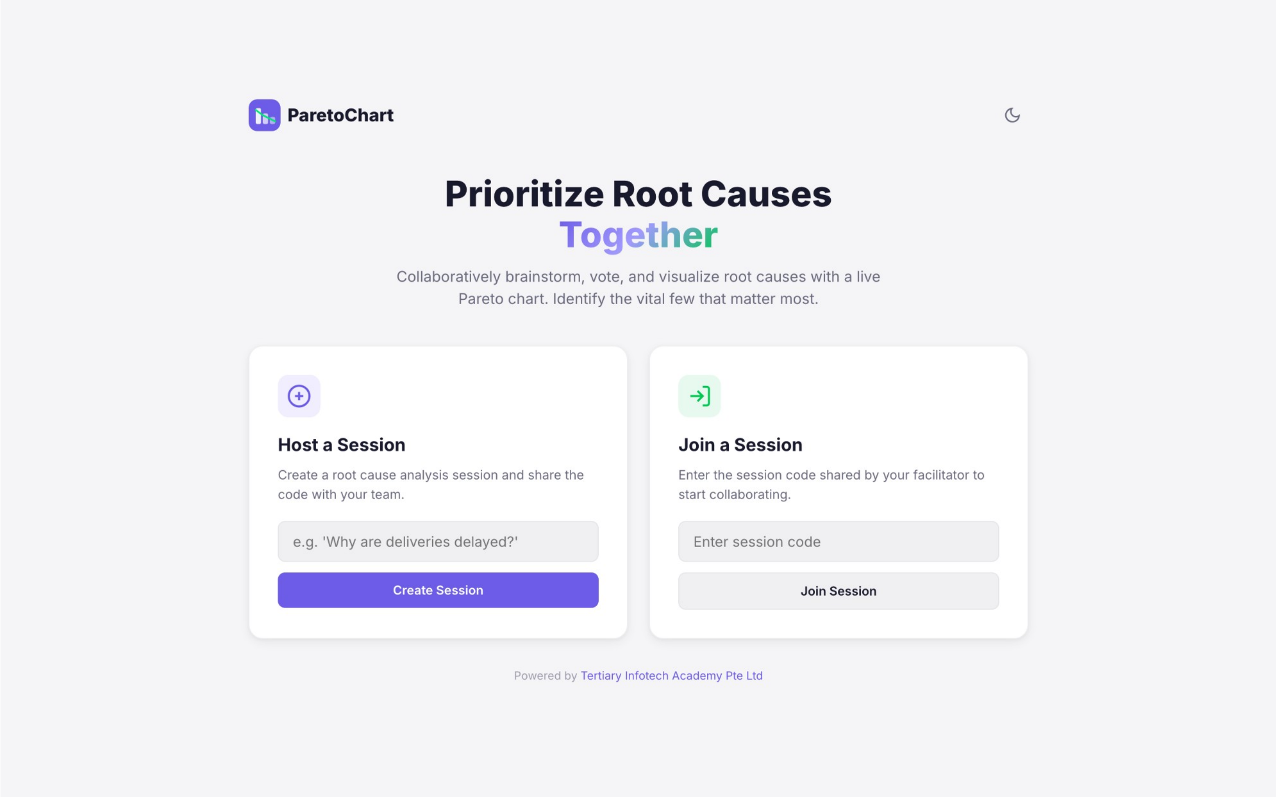
2. A root cause you cannot act on is not a root cause — it is an excuse with a diagram.

3. This is A6 in practice: assessing work performance and quality to drive continuous improvement, with PDCA closing the loop in the next sprint.

SUCCESS CRITERION

Your chain reaches a process or system cause the team can change, not a person and not an external party. Reading it backwards with 'because' holds together at every link. Your SMART action has an owner, a measure and points in the Sprint 2 backlog. If your root cause is 'the carrier's fault', you stopped at Why 1.

The Tool — Pareto Chart



Activity 7 · 45 minutes · full step-by-step: Learner Guide, Activity 7

Open it in your browser

https://alfredang.github.io/paretochart/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

ACTIVITY 7 · 45 MIN

Prioritise Defect Causes with a Pareto Chart

THE SITUATION

Three sprints in, CustomerConnect has accumulated 120 logged defects. The team has capacity to attack roughly two root causes in Sprint 4, and the operations manager wants all 120 fixed. The Scrum Master has categorised every defect by cause and wants the team to decide with data rather than by whoever complains loudest.

WHAT YOU'LL DO

Your team loads the 120-defect dataset into the Pareto tool, builds the ranked bar chart with the cumulative percentage line, identifies the vital few causes crossing the 80% threshold, and converts that finding into a Sprint 4 commitment with a measurable target. You then confront the trade-off with the operations manager.

YOU'LL PRODUCE

A completed Pareto chart of 8 defect categories with a cumulative line, the identified vital few, and a Sprint 4 improvement commitment with a numeric target and the projected defect reduction.

DONE WHEN

Your chart is sorted descending, the cumulative line reaches 100%, and your vital few are justified by the crossing point rather than chosen by intuition. Your Sprint 4 target is numeric and verifiable. If your recommendation is 'fix everything', you have not used the chart to make a decision.

ACT 7

Tool: Pareto Chart — https://alfredang.github.io/paretochart/

LO 3 · Measure progress against targets and reduce waste and defects by identifying the few causes driving most of the failures (A3, A4, A6). · Full step-by-step: Learner Guide, Activity 7

Debrief — Activity 7: What It Proves

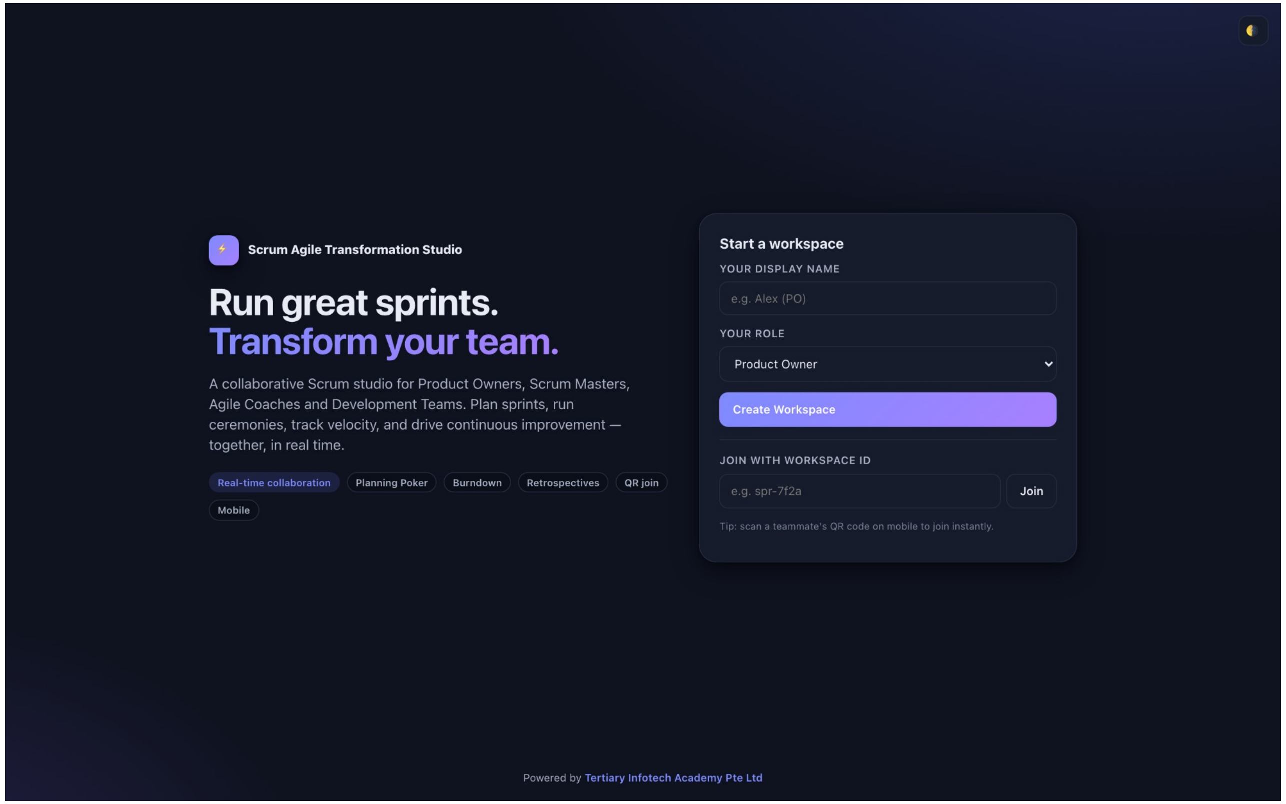
1. Pareto turns 'everything is broken' into 'two things are broken and they cause most of it'.
2. Activities 6 and 7 combine: 5 Whys finds the cause, Pareto proves it is the one worth fixing first.
3. This is A3 and A4 in practice: reducing waste and defects, and measuring against a defined target rather than an impression.

SUCCESS CRITERION

Your chart is sorted descending, the cumulative line reaches 100%, and your vital few are justified by the crossing point rather than chosen by intuition. Your Sprint 4 target is numeric and verifiable. If your recommendation is 'fix everything', you have not used the chart to make a decision.

ACTIVITY 8 · TOOL

The Tool — Scrum Board



Activity 8 · 45 minutes · full step-by-step: Learner Guide, Activity 8

Open it in your browser

https://alfredang.github.io/scrum/

No install, no login

It runs in the browser. Work on one shared screen as a team.

Build the artefact

The deliverable named on the next slide.

Export it

Screenshot or export into your Activity worksheet.

TOPIC 03 · HANDS-ON ACTIVITY

ACTIVITY 8 · 45 MIN

Forecast the Release from Velocity and Read the Agile Metrics

THE SITUATION

The HarbourFront committee meets on Friday and wants one answer: when will the remaining CustomerConnect scope be delivered? The team has completed 4 sprints with velocities of 14, 18, 17 and 21 points. There are 186 points left in the backlog. The programme director wants a single date. The operations manager has separately asked why the support queue keeps growing despite the team 'going faster'.

WHAT YOU'LL DO

Your team computes average velocity, builds a forecast range rather than a single date, plots a release burndown, and interprets a supplied cumulative flow diagram and control chart to locate the bottleneck behind the growing support queue. You then write the committee answer — a range with its assumptions stated.

YOU'LL PRODUCE

A velocity-based forecast range in sprints and calendar dates, a release burndown, a written interpretation of the CFD and control chart naming the bottleneck, and a one-paragraph answer to the committee.

DONE WHEN

Your answer is a range with assumptions, not a single date. You identify Testing as the bottleneck from the CFD and confirm it with the control chart. You explain the growing queue with Little's Law. If your recommendation is 'increase velocity', re-read the control chart — that is what caused the problem.

ACT 8

Tool: Scrum Board — https://alfredang.github.io/scrum/

LO 3 · Measure progress against targets for defined business outcomes and forecast delivery from empirical data (A4, A6, A7). · Full step-by-step: Learner Guide, Activity 8

TOPIC 03 · ACTIVITY 8 DEBRIEF

Debrief — Activity 8: What It Proves

1. Five metrics, one story: burndown for the sprint, release burndown for the release, velocity for the forecast, control chart for cycle time, CFD for the bottleneck.

2. Velocity up and cycle time up at the same time always means WIP is up. Little's Law is not optional arithmetic.

3. This is A4 in practice: measuring progress against targets for defined business outcomes — and being honest with the committee about uncertainty.

SUCCESS CRITERION

Your answer is a range with assumptions, not a single date. You identify Testing as the bottleneck from the CFD and confirm it with the control chart. You explain the growing queue with Little's Law. If your recommendation is 'increase velocity', re-read the control chart — that is what caused the problem.

Recap — Agile Project Execution and Tracking

1. **Activity 5 · Execute Sprint 1 and Track It on the Scrum Board**
   Measure progress against targets on a regular basis by executing a sprint on a live board and reading the burndown (K5, A4, A7).

2. **Activity 6 · Run the Sprint Retrospective with 5 Whys Root-Cause Analysis**
   Assess work performance and quality to ensure continuous improvement, drilling to the root cause of a delivery problem (A6, A7).

3. **Activity 7 · Prioritise Defect Causes with a Pareto Chart**
   Measure progress against targets and reduce waste and defects by identifying the few causes driving most of the failures (A3, A4, A6).

4. **Activity 8 · Forecast the Release from Velocity and Read the Agile Metrics**
   Measure progress against targets for defined business outcomes and forecast delivery from empirical data (A4, A6, A7).

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES ACHIEVED

What You Can Now Do

1

LO1 — Adopt the Agile mindset

You can explain why waterfall concentrates risk, diagnose a failed delivery, and reframe requirements around customer value.

2

LO2 — Share and implement practices

You can apply the Manifesto, run Scrum, build an ordered backlog, plan a sprint and clarify accountability.

3

LO3 — Build, execute and track

You can run a sprint, read all five Agile metrics, find a root cause and forecast a release honestly.

THE TAKEAWAYS

The Eight Things Worth Remembering

1 Fixed time, variable scope

That inversion is the whole paradigm shift.

2 Value early beats value complete

Shipping 20% of scope in month two changes the return profile.

3 The sprint goal makes a sprint

Without one you have a task list with a deadline.

4 Done means Done

Carryover is never velocity, and 'nearly' is not a status.

5 One Accountable per decision

Two Product Owners is the most common Agile failure.

6 Limit WIP to go faster

Little's Law: cycle time = WIP ÷ throughput.

7 Velocity forecasts, never targets

Reward it and you destroy it.

8 Actions need owners

An improvement with no owner and no capacity is a wish.

IF YOU REMEMBER ONE THING

Agile is not a faster way to build what you were told to build. It is a disciplined way to keep finding out what is actually worth building — and to stop building what is not.

RECOMMENDED COURSES

Where to Go Next

1. **WSQ - Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)**
   www.tertiarycourses.com.sg
2. **WSQ - Design Thinking Course for Businesses**
   www.tertiarycourses.com.sg
3. **WSQ - Effective Project Management for Small Projects**
   www.tertiarycourses.com.sg
4. **WSQ - Innovative Problem Solving with Generative AI (GenAI)**
   www.tertiarycourses.com.sg
5. **WSQ - Mastering Agile Project Management for IT Projects**
   www.tertiarycourses.com.sg

Final Assessment

1. **Written Assessment (WA)**
   Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.

2. **Case Study (CS)**
   Case Study (CS) — an applied business scenario with structured tasks, 1 hour, open book.

3. **Open book**
   Open book: slides, Learner Guide and approved course materials only.

4. **Competency**
   A minimum of 75% attendance is required to be eligible for assessment and funding. Learners must be assessed Competent in both instruments.

Assessment Flow

1. **BRIEFING**
Rules, timing, open-book scope

2. **WRITTEN (WA)**
Short-answer questions · 1 hour

3. **CASE STUDY (CS)**
Applied business scenario · 1 hour

4. **MARKING**
Assessor marks against the key

5. **RESULT & APPEAL**
Competent / NYC · feedback given

Both instruments must be assessed COMPETENT · Open book: slides, Learner Guide and approved materials only · 75% attendance required

You must be assessed Competent in BOTH instruments. Feedback and an appeal route are available at the end.

Certificate & TRAQOM Survey (Mandatory)

https://lms-tms.tertiaryinfotech.com

Log in with your registered email

Course feedback complete the TRAQOM survey

Digital attendance Assessment attendance must be submitted

Certificate download once assessed Competent

1. Log in to lms-tms.tertiaryinfotech.com
2. Complete the mandatory TRAQOM survey
3. Submit the Assessment digital attendance
4. Download your Certificate of Achievement
5. Your OpenCert from SSG follows by email

TRAQOM completion and 75% attendance are both mandatory for WSQ funding.

WE'RE HERE AFTER THE COURSE TOO

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4 **LMS / TMS**
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Thank You!

Agile Project Management for Business

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